XPS! (XPS!) of QA! (QA!) on the Ag(100) surface

Focusing Laboratory Course

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Contents

1 Introduction

The interface of organic semiconductors with metal surfaces is of particular importance within the domain of organic electronics. The aforementioned organic electronics include, for example, **OLED!**s (**OLED!**s), **OPV!**s (**OPV!**s) and **OFET!**s (**OFET!**s).[Koch2007]

One organic molecule that has been extensively studied in the context of organic electronics is **QA!** (**QA!**). **QA!** is an organic, prochiral molecule with a large π-conjugated system, which is known for its use as a dye in printer toners and varnishes.[Hye2012] However, its potential extends beyond the domain of dyes, as previous studies demonstrated the application of **QA!** in organic electronics.[Glowacki2013, DanielGlowacki2012, Wang2016] Specific applications for **QA!** have been identified, including its use in **OLED!**s,[Wang2016a, Min2021, Cunha2018] **OPV!**s[Dunst2017, Sung2017] and **OFET!**s.[Jeon2018, Jeong2017, Kanbur2019, Berg2009] In addition, **QA!** has been shown to form well-defined structures on metal surfaces, such as the Ag(100), the Ag(111) and the Cu(111) surface.[Humberg2020, Humberg2024, Humberg2023, Humberg2024a, W

A common structural motif of **QA!** monolayers on surfaces[Humberg2024, Wagner2014, Trixler2007, Eberle2019] and in its bulk structure[Paulus2007] are 1D molecular chains. The formation of the 1D molecular chains is driven by intermolecular hydrogen bonds.[Humberg2024, Humberg2020]

In particular, **QA!** on the Ag(100) surface forms two different phases, the α-phase and the β-phase. The α-phase consists of the previously described homochiral, parallel lying 1D molecular chains. This leads to a **POL!** (**POL!**) structure for the α-phase. In contrast, the β-phase is characterized by a commensurate phase, which is thermodynamically more stable than the α-phase and forms irreversibly when the sample is heated. The β-phase has fewer hydrogen bonds and stronger adsorbate-substrate interactions. In addition to this, the β-phase has proposed to be heterochiral, necessitating a change of the handedness of the molecules during the phase transition.[Humberg2024, Humberg2020]

The phase transition and the differences between the two phases are of particular interest, as the change in the binding situation of the different atoms are not yet known.

The present study is focused on how the binding situation of **QA!** on the Ag(100) surface is impacted by the interplay between intermolecular interactions via hydrogen bonds and adsorbate-substrate interactions across the two distinct phases. Specifically, the quantity of the hydrogen bonds and the specific atoms that are involved in the adsorbate-surface interactions are of interest. Furthermore, the diversity of chemical environments that exist for the different atoms in the molecule and the subsequent alterations that occur upon phase transition, are of particular relevance.

For this purpose, **XPS!** is used as the primary technique. **XPS!** provides insights into the binding energies of the core level electrons for the different types of atoms present in the adsorbate. Consequently, the binding energies of the core level electrons give information about the binding situation of the different substrate atoms on the Ag(100) surface. However, the details about the binding situation of the adsorbate are not accessible with the previously used techniques, such as **STM!** (**STM!**) and **LEED!** (**LEED!**), as they only provide information about the molecular arrangement.

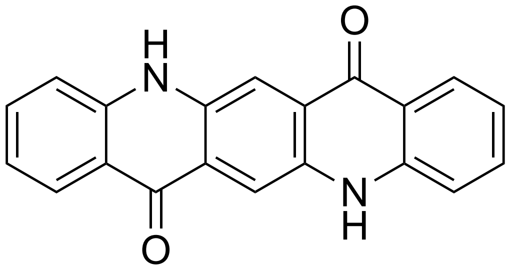
The **XPS!** spectra offer new insights regarding the priorly mentioned binding situation. Previous **STM!** and **LEED!** studies proposed 2 hydrogen bonds per molecule in the α-phase and 1.5 hydrogen bonds per molecule in the β-phase.[**Humberg2024**, **Humberg2020**] Similarly, the present study determined 2 hydrogen bonds per molecule in the α-phase, but in contrast to previous studies, only 1 hydrogen bond per molecule in the β-phase. This finding necessitates a new structure model, a subject that will be examined in the subsequent chapters.

2 Literature

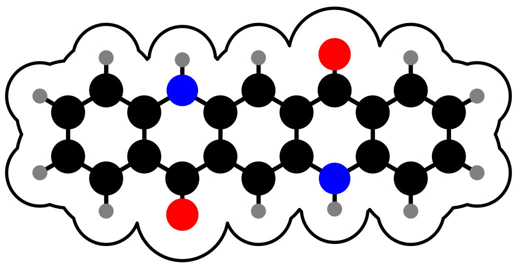
2.1 Quinachridone (QA)

The examined adsorbate **QA!** (**QA!**) is a high symmetric, prochiral organic molecule with the IUPAC name 5,12-dihydro-quino[2,3-*b*]acridine-7,14-dione and the sum formula $C_{20}H_{12}N_2O_2$. Organic crystals exhibit a color which can range from red to violet.[ChemicalBook2025, ChemSpider2025]

QA! is most prominently recognized as the parent structure of a group of pigments, most notably Pigment Violet 19 (C.I. 73900). This particular pigment is extensively utilized in industrial applications due to its remarkable color properties and stability.[ChemicalBook2025, ChemSpider2025] The molecular structure of **QA!** is shown in **??**.



(a) Skeletal structure



(b) Ball-and-stick model

Figure 2.1: Molecular structure of **QA!** (**QA!**).

The molecule is classified as a heterocyclic compound. Its structure is characterized by a conjugated ring system that induces significant rigidity and planarity to the molecule.[forBiotechnologyInformation2025] **QA!** is of the symmetry group C_{2h} and has a molar mass of 312.32 g/mol. The molecule is a crystalline solid with four known different crystal structures under standard conditions.[Paulus2007, Mizuguchi]

2.2 Structures of Quinachridone (QA) on the Ag(100) surface

For **QA!** deposited onto the Ag(100) surface, two structural phases were observed to form: the α- and the β-phase. A thorough examination of these two phases has been conducted by N. Humberg.[Humberg2024, Humberg2020]

The formation of the α-phase occurs when the **QA!** molecules are deposited onto the Ag(100) surface at a sample temperature of 300 K. This phase is composed of parallel molecular chains that are flat-lying with the π-system being oriented parallel to the Ag(100) surface. The **QA!** molecules are connected via intermolecular hydrogen bonds between the keto groups of one molecule and the amine groups of another molecule, resulting in a total of 4 hydrogen bonds. It was determined that the domains under consideration are homochiral, a finding that suggests a high degree of mobility for the **QA!** molecules on Ag(100) surface at 300 K.

In addition, the superstructure matrix of the α-phase for a complete **ML!** (**ML!**) was determined using a **SPA-LEED!** (**SPA-LEED!**) pattern and **STM!** results. The following superstructure matrix was found

$$\mathbf{M}_\alpha = \begin{pmatrix} 2 & 1.25 \\ -3 & 4.80 \end{pmatrix},$$

which shows that the structure is of the **POL!** type. This means that every lattice point of the superstructure falls on a substrate lattice line of the [10] direction of the Ag(100) surface. It should be noted that the intermolecular distance b_1 between the molecules of a chain are determined by the intermolecular hydrogen bonds and does not change with the coverage. The distance between two molecular chains b_2 increases with decreasing coverage.

The **STM!** images, the **SPA-LEED!** pattern and the current structure model of **QA!** on the Ag(100) surface in the α-phase is illustrated in **??**.

The phase transition from the α-phase to the β-phase occurs when the sample is heated up to 500 K for 15 minutes. The phase transition is irreversible. This means that the α-phase is metastable and is stabilized by a kinetic barrier. This kinetic barrier corresponds to the energy required to break the hydrogen bonds between the **QA!** molecules in the α-phase.

For the β-phase, the investigation revealed a 2D network of molecular chains, which are comprised of dimers and manifest periodic indents. The current structure model proposes a single hydrogen bond between the dimers instead of theoretical two possible hydrogen bonds. The two molecules of the dimer form two hydrogen bonds with each other, resulting in 1.5 hydrogen bonds per molecule.

Once more, employing a **SPA-LEED!** pattern and **STM!** results, the superstructure matrix of the β-phase for a complete **ML!** was determined as

$$\mathbf{M}_\beta = \begin{pmatrix} 4 & 3 \\ -5 & 3 \end{pmatrix},$$

which means that the structure is commensurate. In comparison to the **POL!** type of the α-phase, the commensurate structure of the β-phase indicates that

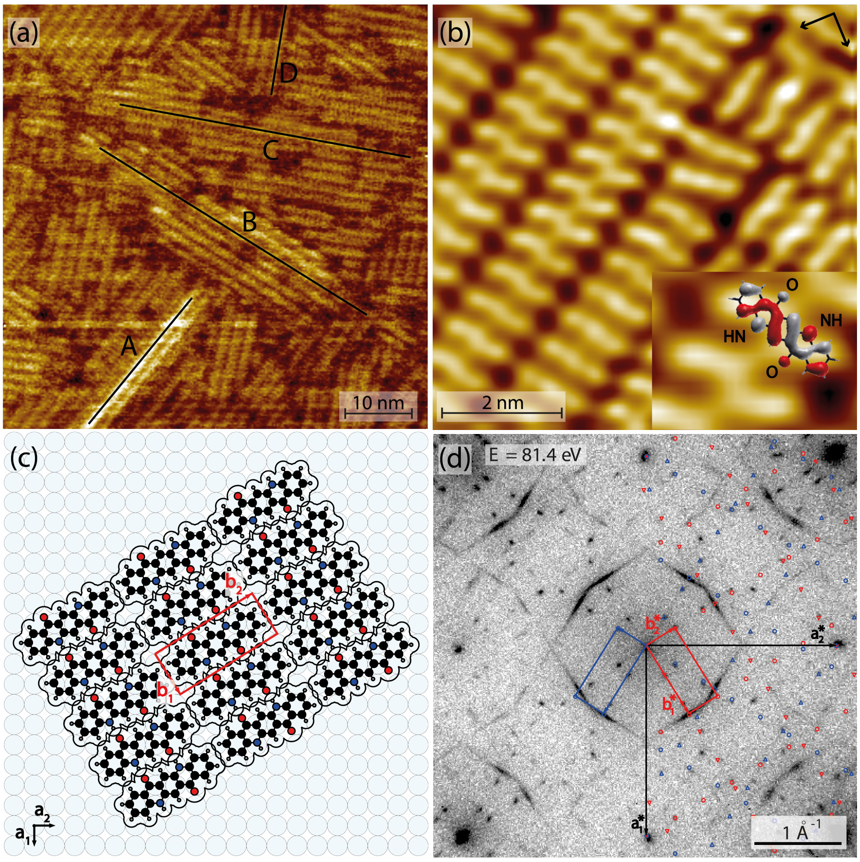


Figure 2.2: Current structure of complete **ML!** of **QA!** on Ag(100) in the α-phase. The image shows an overview **STM!** image (a), a close-up **STM!** image of the molecular arrangement (b), the **SPA-LEED!** pattern (c) and the structure model (d). The image is taken from reference [**Humberg2024**].

the superstructure for the β-phase is aligned with the substrate in a way that the periodicity of the superstructure is a multiple of the substrate lattice constant. In general, the commensurate structure of the β-phase indicates a stronger interaction with the substrate and provides energy gain compared to the α-phase.[**Hooks2001**]

The **STM!** images, the **SPA-LEED!** pattern and the current structure model of **QA!** on the Ag(100) surface in the β-phase is illustrated in **??**.

The current structure model proposed by N. Humberg[**Humberg2020**, **Humberg2024**] already allow for the formulation of a prediction of the **XPS!** spectra. This prediction is based on the different molecular arrangements and interactions in the α-phase and β-phase. In sum, the adsorbate-substrate interaction in the β-phase should be stronger than in the α-phase, leading to lower **BE!**s (**BE!**s) for all components in the β-phase.

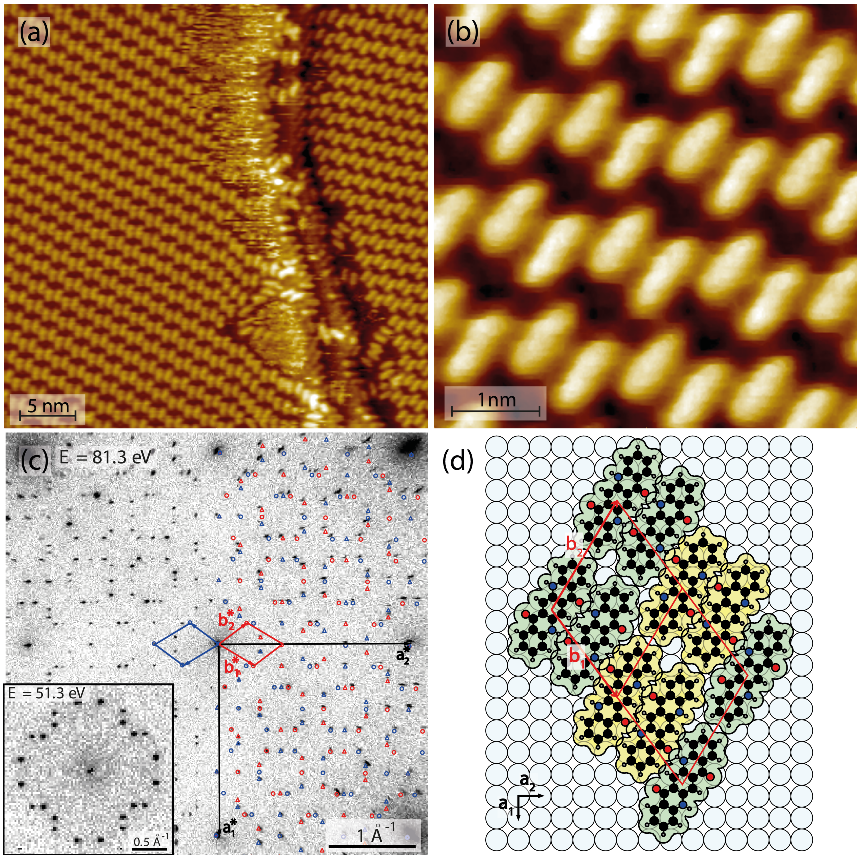


Figure 2.3: Current structure of complete **ML!** of **QA!** on Ag(100) in the β-phase. The image shows an overview **STM!** image (a), a close-up **STM!** image of the molecular arrangement (b), the **SPA-LEED!** image (c) and the structure model (d). The image is taken from reference [**Humberg2024**].

Furthermore, 3/4 oxygen and nitrogen atoms in the β-phase are involved in hydrogen bonds, while in the α-phase all oxygen and nitrogen atoms are involved in hydrogen bonds. This should result in one peak for the oxygen and nitrogen atoms in the α-phase and two distinct peaks with an area ratio of $3 : 1$ for the oxygen and nitrogen atoms in the β-phase.

3 Theory

3.1 X-ray photoelectron spectroscopy (XPS)

XPS! (**XPS!**) utilizes the ejection of electrons from bound states to the continuum by photon excitation, called the photoelectric effect.[**Hertz1887**, **Einstein1905**] The photon energies ($E_\gamma = h\nu$) of x-rays are similar to the **BE!** (E_B) of deep core electrons and are therefore used for the excitation. The experiments for this work were performed with soft x-rays with an energy between 100 - 2100 eV. For each core level, E_γ is chosen to be slightly higher than E_B, resulting in the emission of photoelectrons with a kinetic energy (E_kin) of approximately 100 eV. This leads to a high surface sensitivity of the method, as the mean free path of these electrons is only a few nanometers.[**Fairley2021**]

The photoemission process is illustrated in **??**. The x-ray photons are absorbed by an electron in the core levels of the atom, which results in the ejection of a photoelectron. The energy of the emitted photoelectron is determined by the difference between the photon energy, which is tunable, and the binding energy of the electron in its initial state. Subsequently, the energy of the emitted photoelectron is detected and analyzed to provide information about the change of energy due to the bonding environment and its oxidation state. This phenomenon is referred to as chemical shift.[**Kolasinski2012**]

The kinetic energy of the photoelectron measured by the analyzer is relative to the Fermi level E_F and given by:

$$E'_\mathrm{kin} = E_\mathrm{kin} + \phi_\mathrm{sample}, \tag{3.1}$$

where E_kin is the kinetic energy of the photoelectron and ϕ_sample is the work function of the sample. The kinetic energy of the photoelectron is defined as:

$$E_\mathrm{kin} = E_\gamma - E_\mathrm{B} - \phi_\mathrm{sample}, \tag{3.2}$$

where $E_\gamma = h\nu$ is the photon energy and E_B is the binding energy of the electron.

E_B of an electron is defined as the difference in energies between the atom with n electrons and the ion with $n - 1$ electrons:

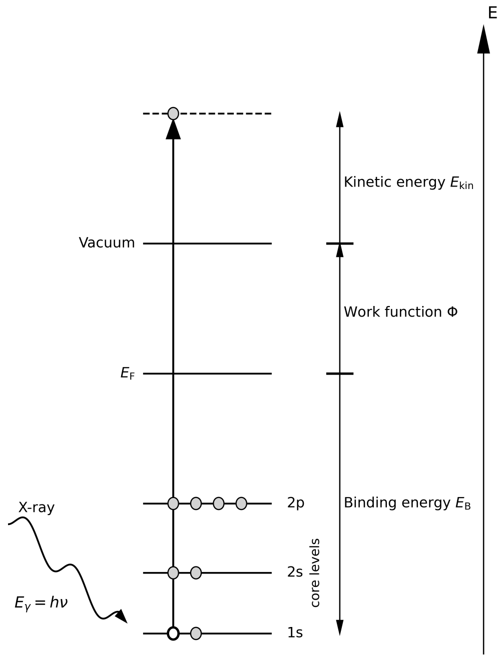


Figure 3.1: Schematic drawing of the photoemission process of an electron in the 1s orbital upon x-ray excitation. The image was adapted using reference [**Attard1998**].

$$E_B = E_f(n-1) - E_i(n), \tag{3.3}$$

As seen in **??**, E_B of photoelectrons depends on initial and final state effects.

The initial state effects manifest for the neutral unexcited atom, due to its interaction with its immediate electronic environment, also referred to as the chemical environment. In addition, final state effects emerge from the correlation between the photoemission process and the ionized final state. The rapid nature of the photoemission process may preclude the system from attaining adiabatic equilibrium.[**Woodruff2016**]

The final state effects are characterized by the presence of satellites with reduced kinetic energy, correlating to increased E_B. This is exemplified by phenomena such as shake-up lines or asymmetric line shapes of the signal, which are indicative of photoemission from the metal surface. The asymmetric line shapes result from coupling of the photoelectrons to the conduction electrons.[**Moulder1993**] A multitude of factors have been identified as contributors to the line shape, particulary the **FWHM!** (**FWHM!**), of the photoemission lines. These include phonon broadening, the lifetime of the photohole, the spectral resolution of the exciting x-ray beam and the energy resolution of the analyzer.[**Moulder1993**, **Fairley2021**]

4 Experimental

4.1 Material

4.1.1 The Ag(100) crystal surface

A silver (Ag) single crystal that was provided by the team of the I09 beamline of the Diamond Light Source with a surface orientation of (100) was used as the substrate for the **XPS!** experiments. It is important that a single crystal is used, as this has a uniform and precisely defined surface and has almost no structural defects.

Silver forms a **fcc!** (**fcc!**) crystal system, which is shown in **??**. This corresponds to the Pearson symbol cF. In the crystal system of silver, one silver atom is surrounded by twelve next neighbor atoms. Furthermore, the Ag(100) surface does not reconstruct. The lattice constant of silver in the volume crystal is 4.0853±0.0013 Å at a temperature of 23 ± 3 °C.[Liu1973]

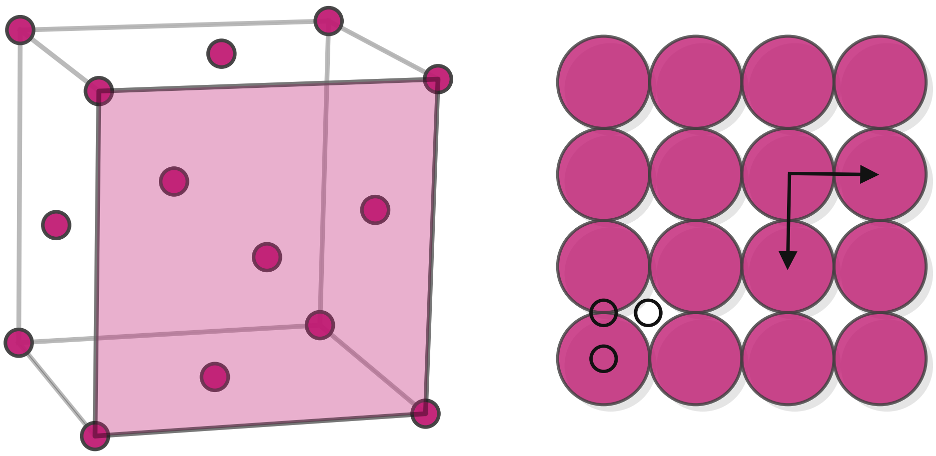


Figure 4.1: Unit cell of the silver crystal with marked (100) plane and the Ag(100) surface with the unit cell vectors (black arrows) and the distinct adsorption sites (black circles).

As seen in **??**, the unit cell of the Ag(100) surface is quadratic with a lattice constant $a_1 = a_2 = 4.0853$ Å$/\sqrt{2}$ =2.8887 Å. The Ag(100) surface has different adsorption sites called fcc, on-top and bridge. These sites are indicated by the black circles in **??**.

4.2 Experimental setup

All **XPS!** measurements were performed in an **UHV!** (**UHV!**) chamber at the I09 endstation at the Diamond Light Source in Didcot, UK. A schematic overview of the I09 endstation is depicted in **??**.

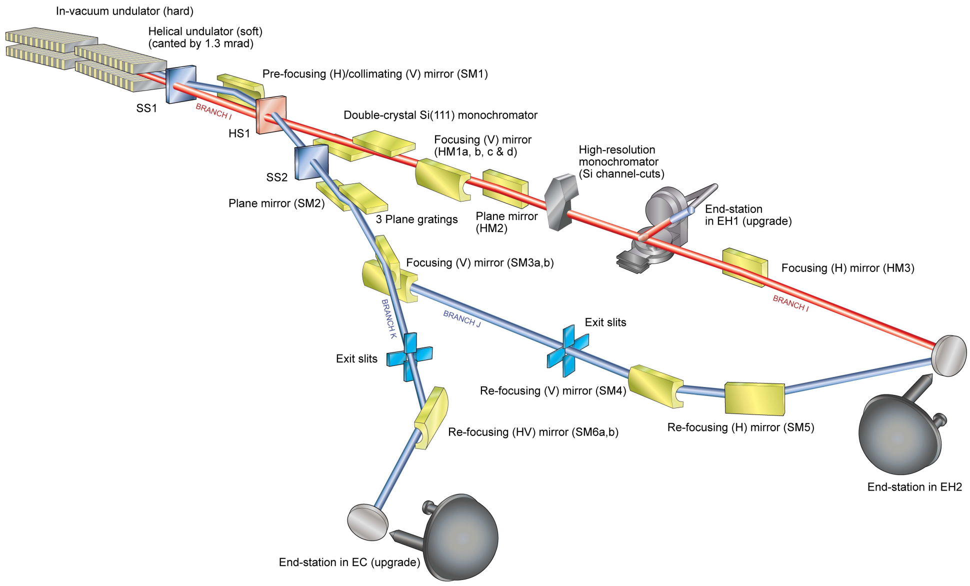


Figure 4.2: Schematic drawing of the I09 endstation of the Diamond Light Source in Didcot, UK. Picture taken from reference [**Diamond2025**].

This particular endstation for the reported experiments (in EH2, **??**) utilizes radiation from an undulator located inside the electron storage ring of the synchrotron. The I09 endstation is equipped with a nitrogen-cooled Si(111) double crystal monochromator (hard x-rays) and a grating monochromator (soft x-rays). This setup enables high-intensity and precise x-ray measurements in the soft and hard x-ray ranges. The emitted photoelectrons were detected by a hemispherical EW4000 HAXPES analyzer purchased from Scienta Omicron. The analyzer has an acceptance cone of 56° and was mounted in the photon polarization plane.[Diamond2025] A schematic drawing of the sample position for the **XPS!** measurements is illustrated in **??**.

The experimental sample preparation is described in the following. First, the Ag(100) surface was cleaned by sputtering with Ar^+ ions. Therefore, argon was dosed into the chamber up to a pressure of $5 \cdot 10^{-5}$ mbar and Ar^+ ions were accelerated with a voltage of 1 keV. Afterwards, the surface was heated up to 825 K for 45 min to increase the thermal diffusion and heal defects on the Ag(100) surface.

Subsequently, **QA!** was evaporated on the Ag(100) surface with the sample being at room temperature. Therefore, the evaporator was heated to about 500 °C. The

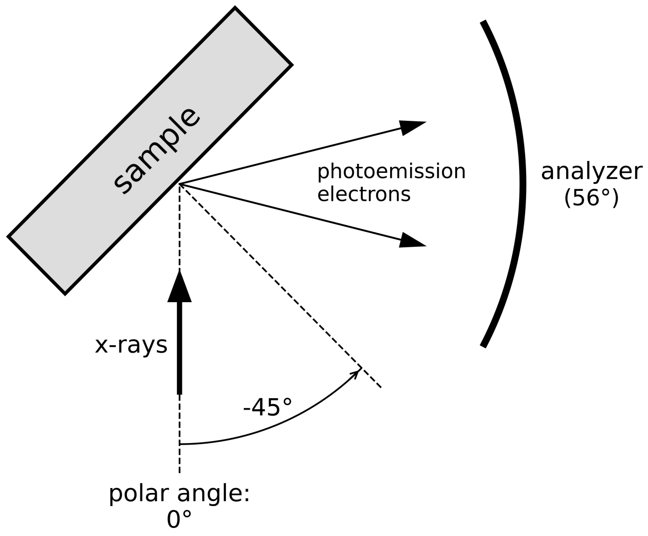


Figure 4.3: Top view of the sample geometries for the **XPS!** measurements at the I09 endstation of the Diamond Light Source in Didcot, UK. Illustrated as in reference [**Kny2025**].

deposition process was performed according to procedures derived in the lab in Bonn, where the depostion could be controlled by a mass spectrometer and **SPA-LEED!** measurements.

After evaporation, the phase of the **QA!** molecules on the Ag(100) surface was characterized using **LEED!**. Exemplary **LEED!** pattern of the α- and the β-phase are shown in **??**. The α-phase shows a **POL!** type superstructure with its characteristic streaky spots in the diffraction pattern, while the β-phase is commensurate with the Ag(100) surface and exhibits sharp spots in the diffraction pattern.

For the measurements of the **XPS!** spectra, x-ray radiation with an energy of either 500 eV for C1s and N1s or 630 eV for O1s was used. For data collection, the sample normal was rotated by -45° out of the x-ray beam in the polar direction towards the analyzer.

The detailed parameters for the data acquistion are listed in **??**. The pass energy E_{pass} of the analyzer determines the kinetic energy of the photoelectrons and therefore the energy resolution of the measurement. The step width is the energy difference between two consecutive data points in the **XPS!** spectrum. The number of frames is the number of times the same spectrum was measured and averaged to reduce noise in the data.

Table 4.1: Experimental parameters for **XPS!** measurements at the I09 endstation of the Diamond Light Source in Didcot, UK.

	photon energy E_γ	pass energy E_{pass}	step width	number of frames
C1s	500 eV	20 eV	40 meV	17
O1s	630 eV	20 eV	40 meV	17
N1s	500 eV	20 eV	40 meV	17

The parameters for the measurements were chosen to achieve a good signal-to-noise ratio and a high energy resolution. Furthermore, the acquisition time for each spectrum was as short as possible to prevent beam damage of the sample.

XPS! spectra for the Fermi energy E_{F} were measured after each **XPS!** measurement of C1s, O1s and N1s for calibrating the **BE!** of the measured spectra against the Fermi level. E_{F} was measured with the same photon energy as the corresponding orbital and a pass energy of 20 eV. The step width was set to 40 meV and the number of frames was changed for each measurement.

4.3 Data processing

The measured data were then subjected to an initial processing stage, whereby the values from multiple datasets employing identical settings for the measurements were averaged to obtain an increased signal-to-noise ratio. The number of averaged spectra was different for the various preparations. Subsequently, the data underwent further processing with the software CasaXPS.[**CasaSoftwareLtd2022**] Each XPS spectrum was processed in a consistent manner, as outlined below.

Initially, the **BE!** of the **XPS!** spectrum was calibrated against the Fermi energy E_{F}. For this purpose, an **XPS!** spectrum of the Fermi edge with the same photon energy as the corresponding component was fitted with a Fermi function. Without inaccuracies of the devices, E_{F} should be zero. Accordingly, the binding energy was shifted by the value of the Fermi energy, which was determined from the fit of the Fermi function.

Afterwards, a linear background was fitted to the averaged and calibrated data. This background was then substracted for further data processing. Then, the corresponding fitting models from **??** were applied. The different peaks with line shape and constraints from **??** were used. These settings were the same for each processed dataset. Subsequently, the processed data was exported from CasaXPS and illustrated using a customized python script, enabling the generation of various plots as outlined in **??**.

Table 4.2: Parameters used in CasaXPS**CasaSoftwareLtd2022** for the different peaks of the **XPS!** spectra of **QA!** on Ag(100).

peak	line shape	area ratio
C_{arom}	LA(1,8,800)	10
C_{NH}	LA(1,8,800)	4
C_{CO}	LA(1,8,800)	4
C_{O}	LA(1,8,200)	2
O1	LA(1,8,400)	1 (for β-phase)
$O1_{sat}$	LA(1,1,400)	-
O2	LA(1,8,400)	1 (for β-phase)
$O2_{sat}$	LA(1,1,400)	-
N1	LA(1,8,400)	1 (for β-phase)
$N1_{damage}$	LA(1,8,400)	-
N2	LA(1,8,400)	1 (for β-phase)

For the line shapes, the LA(α,β,m) function was used, which is a numerical convolution of a Lorentzian and a Gaussian. The parameters α and β define the asymmetry of the line shape, where α describes the left part and β the right part. The parameter m defines the width of the Gaussian.

The line shapes for the different components were slightly changed in order to observe the best description of the **XPS!** spectra. In general, the objective was to maintain consistency in the lines to the greatest extent possible.

Furthermore, the areas of the peaks were constrained to a certain value in order to ensure that the areas of the peaks are consistent with the stochiometry of **QA!**. The area constraints for the C1s **XPS!** spectra were the same for both phases, whereas the area constraints of the O1s and N1s **XPS!** spectra differed between the two phases. This was due to the changed number of hydrogen bonds per molecule in the two phases, which led to a changed number of distinct oxygen and nitrogen atoms in the **XPS!** spectra.

5 Results

The ensuing results will be presented under three topics: the α-phase (**??**), the phase transition from the α- to the β-phase (**??**) and the β-phase (**??**). A compendium of all recorded **XPS!** spectra, organized according to the different photoemission lines, is available for perusal in **??**.

5.1 The α-Phase

The investigation of the α-phase was conducted on all three available photoemission lines: C1s, O1s and N1s. The ensuing discourse will meticulously examine the α-phase.

Therefore, the present chapter has been subdivided into discrete sections, each dedicated to a specific atom type. Within each section, an overview of all recorded **XPS!** spectra for that particular atom type is provided, along with a more detailed view of a representative **XPS!** spectrum for a monolayer. A comparison will be made between the monolayer and the multilayer **XPS!** spectrum.

5.1.1 C1s spectra

?? presents a series of C1s **XPS!** spectra of **QA!** adsorbed on the Ag(100) surface, recorded at varying coverages ranging from 0.08 **ML!** to 1 **ML!**. All spectra exhibit a dominant peak in the region around 285 eV, which is consistent across all coverages. Furthermore, a slight shoulder towards lower **BE!**s is visible. Towards higher **BE!**s, the **XPS!** spectra show a more complex pathway with at least one peak.

Despite the overall similarity in the spectra, particulary in peak position and general shape, there are notable differences in intensity and variations in peak broadening as a function of coverage. The most apparent trend is the systematic increase in signal intensity with increasing coverage, indicating a proportional growth in the number of photoemitting C atoms due to the accumulation of molecules.

The spectrum with the highest integrated intensity that was not a multilayer spectrum is set to 1 **ML!**. The difference between the monolayer and multilayer spectrum was the significant change in the peak shape, as seen in **??**. The adsorbed layers of the other spectra were assigned accordingly using their integrated intensities. This classification of the coverage was used for all **XPS!** spectra for the α-phase. From the C1s **XPS!** spectra, it can be concluded that there are different peaks corresponding to the C atoms in **QA!**. This will be discussed in more detail below.

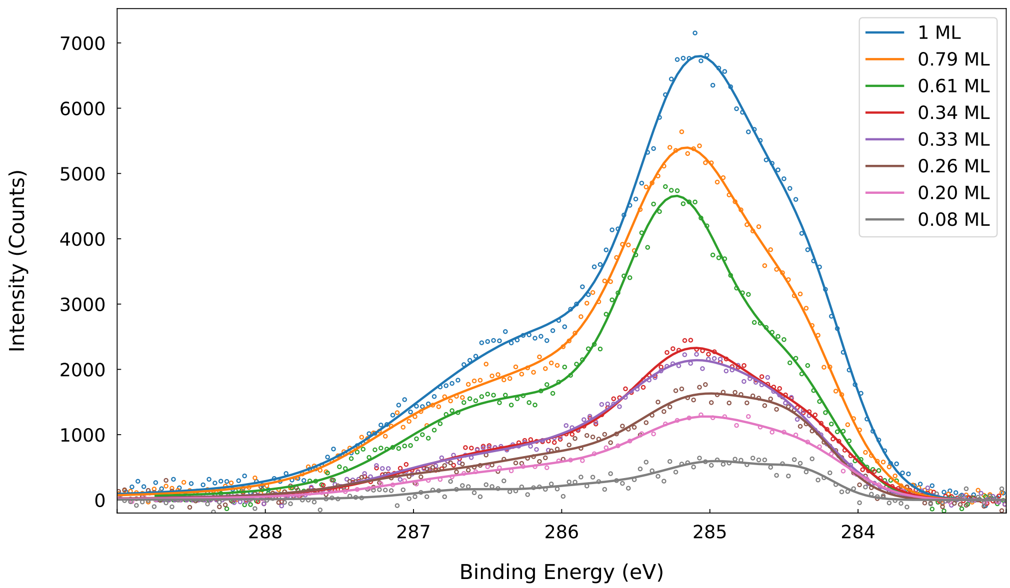


Figure 5.1: Overview of C1s **XPS!** spectra for monolayer preparations of the α-phase of **QA!** on Ag(100) with a coverage between 0.08 **ML!** and 1.00 **ML!**. The lines correspond to the envelope of the fitted peaks and the circles are the measured data points.

The **XPS!** spectrum for a coverage of 1 **ML!** in **??** demonstrates different peaks that present C atoms with different chemical environments in **QA!**.

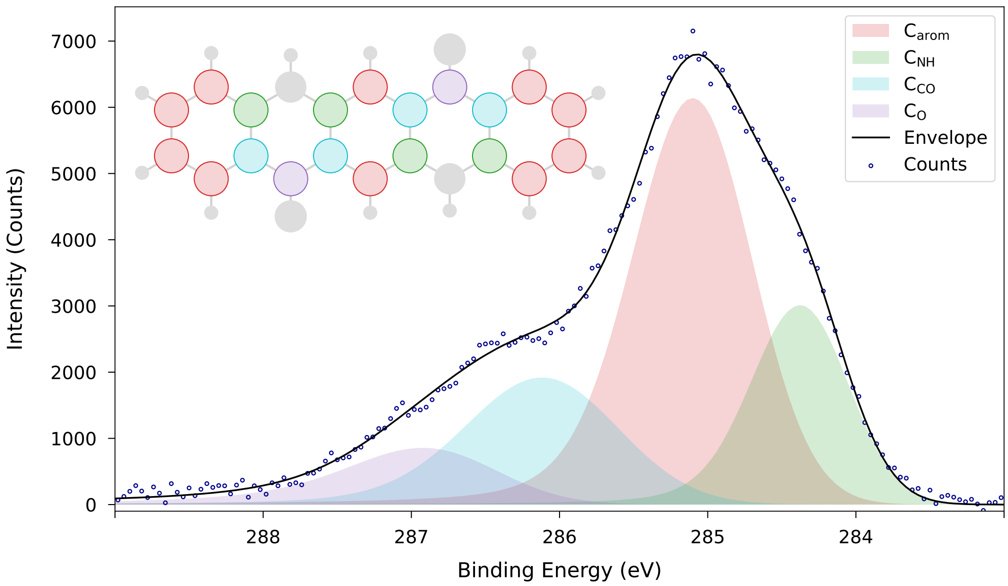


Figure 5.2: C1s **XPS!** spectrum for the α-phase of **QA!** on Ag(100) for a coverage of 1 **ML!** with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line). Four C1s **XPS!** spectra were averaged to obtain this spectrum.

The **XPS!** spectra for the C1s photoemission lines of **QA!** on the Ag(100) surface reveal consistent spectral features. The main peak (C_{arom}) contains contributions of the C atoms from the aromatic rings that are not part of the two rings containing the keto and the amine groups. Next, there is a peak for the neighbor C atoms of the amine (C_{NH}) and another peak for the neighbor C atoms of the ketone (C_{CO}). Finally, there is one peak for the carbonyl C atoms of the ketone (C_{O}). The values corresponding to each peak can be found in **??**.

Table 5.1: Fit parameters used in CasaXPS**CasaSoftwareLtd2022** for the monolayer α-phase of **QA!** on Ag(100) for the C1s photoemission lines.

peak	**BE!** / eV	area ratio	FWHM / eV
C_{arom}	285.02	10	0.94
C_{NH}	284.31	4	0.76
C_{CO}	286.01	4	1.20
C_{O}	286.72	2	1.15

As illustrated in **??** and **??**, the detailed fitting model can be described as follows: The **BE!** of the C_{arom} peak is measured at 285.02 eV, with a **FWHM!** of 0.94 eV. The C_{NH} peak is next to the aforementioned peak, with a **BE!** of 284.31 eV and a **FWHM!** of 0.76 eV. The analysis of the data reveals that the C_{CO} peak is located to the left of the C_{arom} peak. The **BE!** of the C_{CO} peak is 286.01 eV and its **FWHM!** is 1.20 eV. The C_O peak is located to the left of this peak at a **BE!** of 286.72 eV and a **FWHM!** of 1.15 eV.

In contrast to the α-phase C1s **XPS!** spectra of the monolayer, the multilayer spectrum in **??** exhibits a different shape.

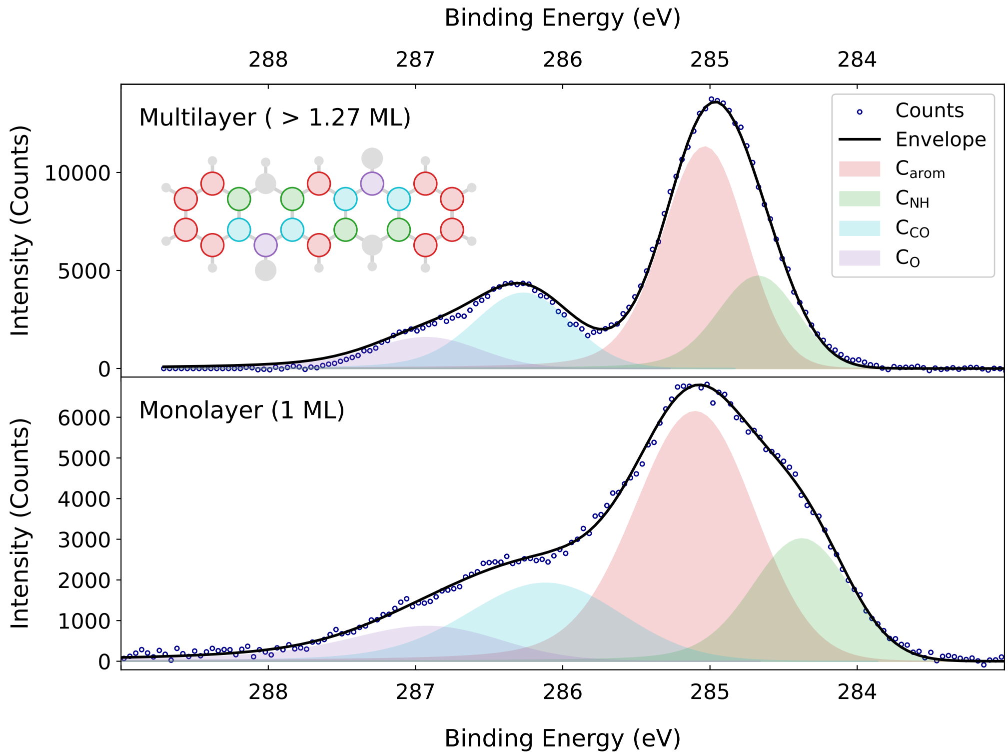


Figure 5.3: C1s **XPS!** spectrum for the α-phase of **QA!** on Ag(100) for a mutlilayer coverage compared to the **XPS!** spectrum for a coverage of 1 **ML!** with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line).

The coverage for the multilayer spectrum in **??** was estimated to be above 1.27 **ML!**. The coverage was determined by the integrated intensity of the C1s multilayer **XPS!** spectrum, which was compared to the integrated intensity of the 1 **ML!** **XPS!** spectrum. Nevertheless, the exact coverage is not known, as the intensity of the photoemission lines of the multilayer is not directly proportional to the number of adsorbed molecules, because the sensitivity of the photoemission lines changes for different layers. This leads to the assumption that the coverage was above 1.27 **ML!**.

Initially, the multilayer spectrum is still adequately characterized by using the four distinct peaks, which possess equivalent area ratios to those of the monolayer spectrum. However, it is evident that the multilayer spectrum exhibits a significantly higher intensity relative to the monolayer spectrum. The primary distinctions are evident in the **BE!** and the **FWHM!** of the peaks. The spectrum displays a discernible minimum between the C_{arom} peak and the C_O peak. The peak associated with the amine group (C_{NH}) exhibits a shift towards the C_{arom} peak.

The parameters for the multilayer spectrum in **??** are listed in **??**.

Table 5.2: Fit parameters used in CasaXPS[**CasaSoftwareLtd2022**] for the multilayer α-phase of **QA!** on Ag(100) for the C1s photoemission lines.

peak	**BE!** / eV	area ratio	FWHM / eV
C_{arom}	284.98	10	0.65
C_{NH}	284.62	4	0.62
C_{CO}	286.20	4	0.76
C_O	286.80	2	0.85

5.1.2 O1s spectra

?? presents a series of **XPS!** spectra for the O1s photoemission lines of **QA!** at three different coverages: 0.33 **ML!**, 0.79 **ML!** and 1 **ML!**. A prominent peak centered near 531 eV is exhibited by all spectra, a characteristic of the O atoms in carbonyl functional groups present in the **QA!** molecule. Additionally, the shape of the peak suggests the presence of a more intricate structure than a single peak. The spectra exhibit a comparable overall shape and peak position across the various surface coverages.

As anticipated, the spectral intensity increases with coverage, indicative of an increased number of **QA!** molecules adsorbed on the surface. Beyond the intensity changes, differences in the peak width and asymmetry are also observed. The spectrum at the lowest coverage (0.33 ML) is narrower and more symmetric, whereas the spectra at higher coverages become progressively broader and more asymmetric.

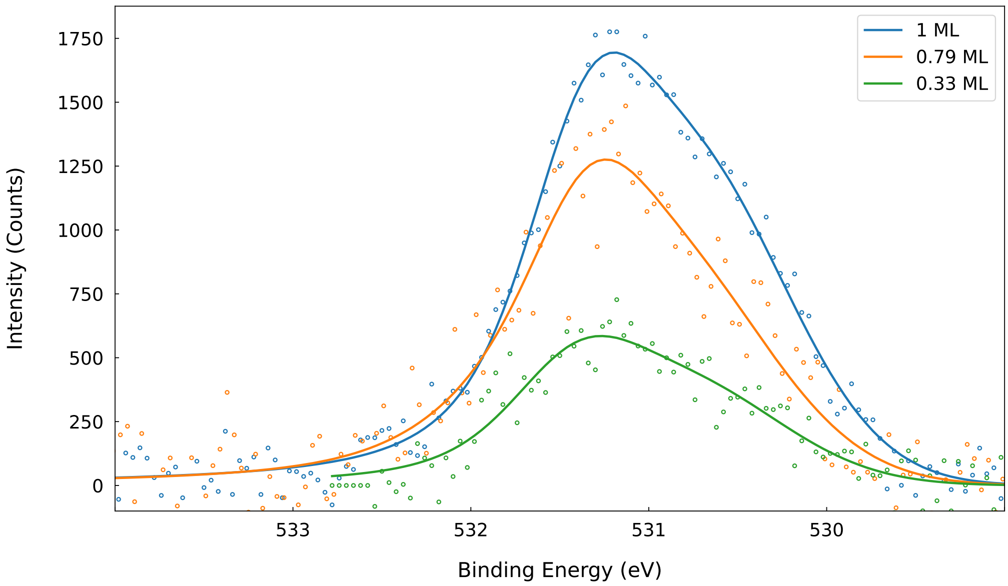


Figure 5.4: Overview of O1s **XPS!** spectra for monolayer preparation of the α-phase of **QA!** on Ag(100) with a coverage between 0.33 **ML!** and 1.00 **ML!**. The line corresponds to the envelope of the fitted peaks and the circles are the measured data points.

The O1s **XPS!** spectrum of **QA!** on Ag(100) at a 1 **ML!** coverage reveals two distinguishable spectral components, which is shown in **??**.

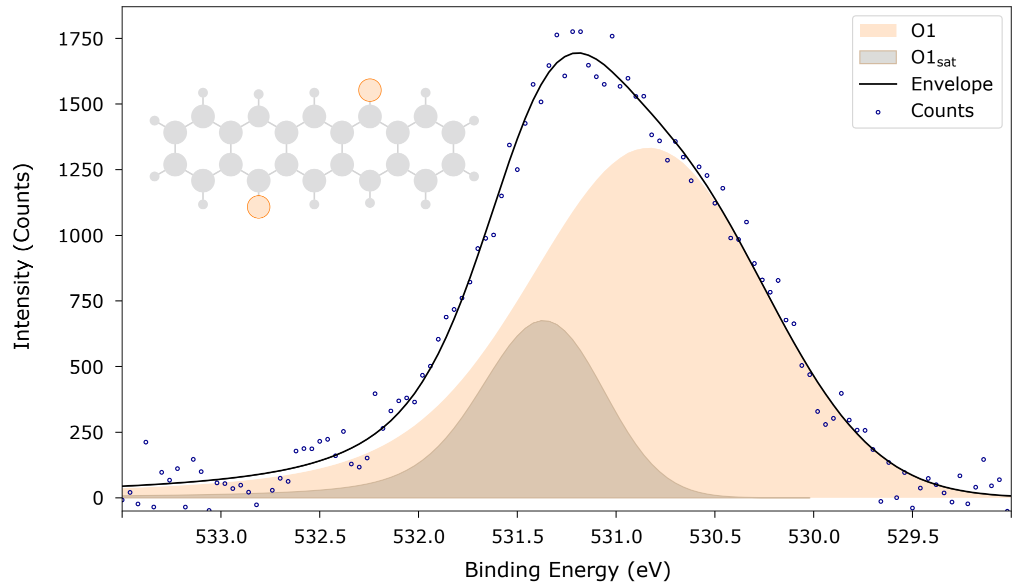


Figure 5.5: O1s **XPS!** spectrum for the α-phase of **QA!** on Ag(100) for a coverage of 1 **ML!** with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line). Five O1s **XPS!** spectra were averaged to obtain this spectrum.

As illustrated in **??** and **??**, the primary component (O1) is at a **BE!** of 530.66 eV, exhibiting a **FWHM!** of 1.36 eV. This peak is attributed to the core-level photoemission from O atoms situated within carbonyl functionalities. The relatively broad linewidth in comparison to the linewidths of the C1s **XPS!** spectra is indicative of slight variations in the local chemical environment of these O atoms.

A secondary, less intense component ($O1_{sat}$) is observed at a higher **BE!** of 531.27 eV with a narrower **FWHM!** of 0.71 eV. This feature is interpreted as a shake-up satellite, arising from e.g. a $\pi - \pi^*$ transition within the conjugated π system of the **QA!** molecule.

Quantitative analysis of the peak areas reveals that the satellite-to-main peak area ratio is approximately 1 : 0.27, which is equivalent to 21 % of the total O1s intensity. This ratio is consistent with the expected intensity of shake-up features in aromatic systems with significant conjugation.[**Bauer2014**]

The combination of these two components provides a comprehensive representation of the O1s photoemission lines of **QA!**. This representation reflects both the intrinsic chemical identity of the O atoms and the dynamic electron processes that accompany core-level photoemission.

Table 5.3: Fit parameters used in CasaXPS[CasaSoftwareLtd2022] for the monolayer α-phase of **QA!** on Ag(100) for the O1s photoemission lines.

peak	**BE!** / eV	area ratio	FWHM / eV
O1	530.66	2.00	1.36
$O1_{sat}$	531.27	0.53	0.71

In **??**, a **XPS!** spectrum with a multilayer coverage is compared to the previously discussed **XPS!** spectrum.

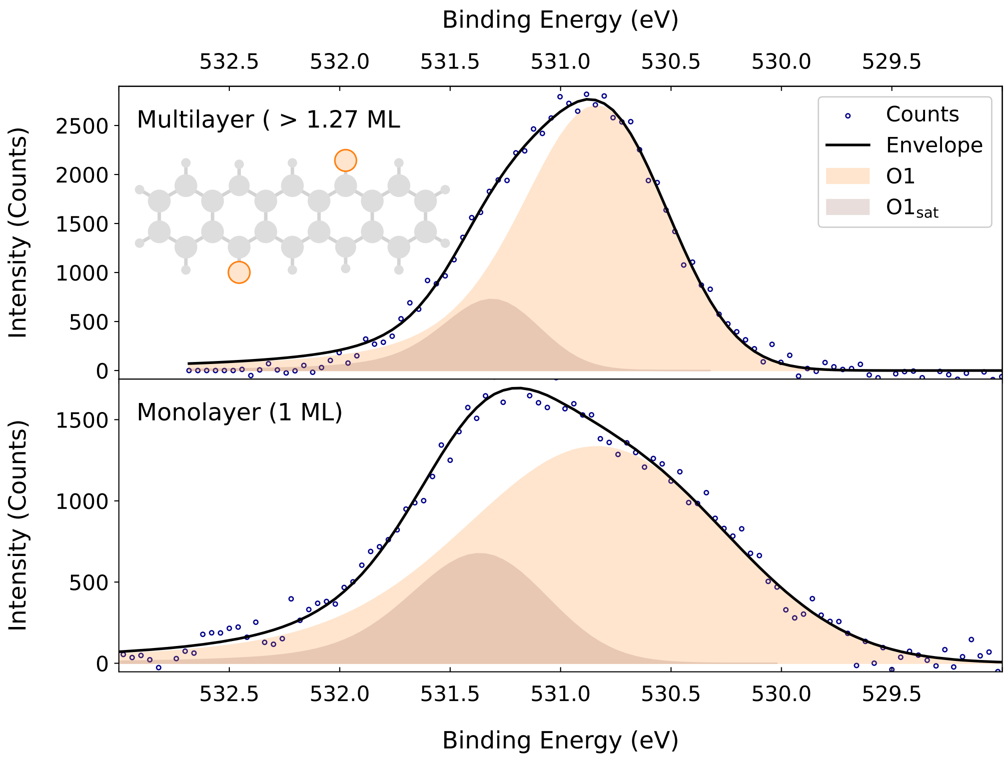


Figure 5.6: O1s **XPS!** spectrum for the α-phase of **QA!** on Ag(100) for a mutlilayer coverage compared to the **XPS!** spectrum for a coverage of 1 **ML!** with the experimental data points, the fitted components and the resulting envelope.

The O1s **XPS!** spectrum recorded for **QA!** on Ag(100) at multilayer coverage reveals a well-defined structure characterized by two distinct photoemission peaks. The primary component (O1) is at a **BE!** of 530.733 eV and a **FWHM!** of 0.75 eV. In comparison with the **ML!** O1 peak, the peak position undergoes a change of approximately 0.076 eV. A more salient finding is the increase in the **FWHM!** of approximately 0.61 eV, which nearly corresponds to a 50 %.

The satellite peak ($O1_{sat}$) is observed at 531.24 eV with a **FWHM!** of 0.5 eV. Once more, the position of the $O1_{sat}$ peak in relation to the **ML!** remains constant. However, the **FWHM!** of the peak decreases by 0.21 eV, which corresponds to approximately 30 %. This alteration may be attributable to the enhanced intensity exhibited by the multilayer **XPS!** spectrum. The satellite-to-main ratio experiences a decrease from 1 : 0.27 in the **ML!** to 1 : 0.18 in the multilayer, indicating a reduced shake-up process.

The parameters for the O1s **XPS!** spectrum with a multilayer coverage in **??** are listed in **??**.

Table 5.4: Fit parameters used in CasaXPS[**CasaSoftwareLtd2022**] for the multilayer α-phase of **QA!** on Ag(100) for the O1s photoemission lines..

peak	**BE!** / eV	area ratio	FWHM / eV
O1	530.73	2.00	0.75
$O1_{sat}$	531.24	0.36	0.50

5.1.3 N1s spectra

The **XPS!** spectra displayed in **??** show the N1s photoemission lines of **QA!** on Ag(100) with varying coverages between 0.26 **ML!** and 1 **ML!**. A prominent main peak centered around 400.1 eV is exhibited by all spectra. Despite the general spectral similarities across different coverages, clear distinctions emerge in the shape and intensity of the main peak. As the coverage increases from submonolayer to monolayer, the main peak systematically increases in intensity, reflecting the proportional increase in nitrogen-containing molecules. In scenarios where coverages are less substantial, the spectra tend to manifest as marginally broader. Conversely, as the coverage increases, the spectra become more narrow.

Additionally, a secondary peak is evident on the low-**BE!** side of the main peak in some spectra at approximately 397.7 eV. In summary, while the core features of the **XPS!** of **QA!** remain largely preserved across different coverages, changes are observed in the peak intensity and the width, in addition to the second peak at a lower **BE!**.

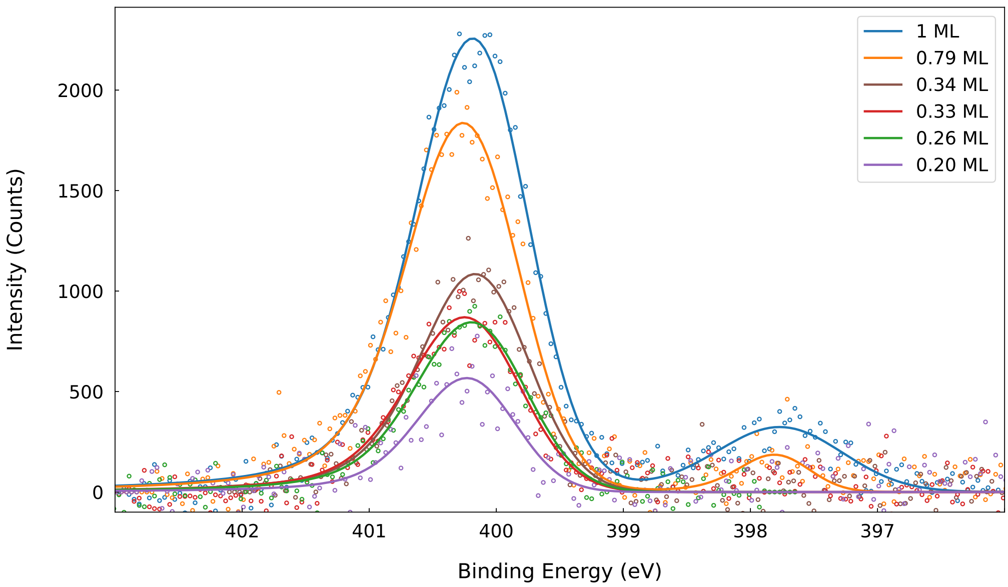


Figure 5.7: Overview of N1s **XPS!** spectra for monolayer preparation of the α-phase of **QA!** on Ag(100) with a coverage between 0.26 **ML!** and 1.00 **ML!**. The line corresponds to the envelope of the fitted peaks and the circles are the measured data points.

The N1s **XPS!** spectrum of **QA!** on Ag(100) with a coverage of 1 **ML!**, as depicted in **??**, is discussed in more detail in **??**, which presents the different peaks.

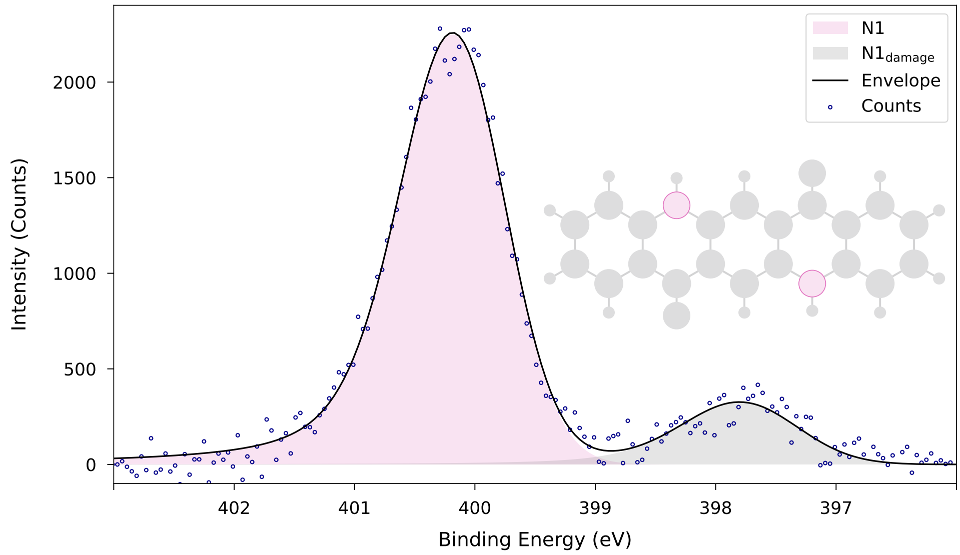


Figure 5.8: N1s **XPS!** spectrum for the α-phase of **QA!** on Ag(100) for a coverage of 1 **ML!** with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line). Four N1s **XPS!** spectra were averaged to obtain this spectrum.

The **XPS!** spectrum in **??** can be divided into two discrete components: a primary peak (N1) and a secondary feature at lower **BE!** ($N1_{damage}$). The precise values for the two peaks are enumerated in **??**. The N1 peak, with a center at 400.05 eV and a **FWHM!** of 1.02 eV, is attributed to N atoms embedded within the chemically intact **QA!** molecule. The narrow **FWHM!** and symmetrical peak profile are indicative of a well-defined and electronically homogenous nitrogen environment.

The secondary component, $N1_{damage}$, emerges at a **BE!** of 397.66 eV and exhibits a **FWHM!** of 1.16 eV. This phenomenon is attributed to the presence of chemically modified nitrogen species, which are most likely the result of radiation-induced damage due to prolonged x-ray exposure. The assumption that the N atoms in **QA!** are susceptible to radiation damage is supported by the **XPS!** spectra for 1 **ML!** and 0.79 **ML!** coverages, which exhibit a different area ratio of the main component to the damage component. These two **XPS!** spectra were longer exposed to the x-ray radiation than the other **XPS!** spectra without a second peak, explaining the observed peak due to beam damage.

A thorough qualitative analysis was conducted, which yielded a damage-intensity-ratio of 0.16 for the 1 **ML!** coverage. This ratio indicated that the majority of N atoms maintain their molecular integrity under the measurement conditions, while a minor but significant portion experiences beam damage.

Table 5.5: Fit parameters used in CasaXPS**CasaSoftwareLtd2022** for the monolayer α-phase of **QA!** on Ag(100) for the N1s photoemission lines.

peak	**BE!** / eV	area ratio	FWHM / eV
N1	400.05	2.00	1.02
$N1_{damage}$	397.66	0.31	1.16

A comparison of the N1s **XPS!** spectrum of the monolayer and the spectrum corresponding to the multilayer is illustrated in **??**.

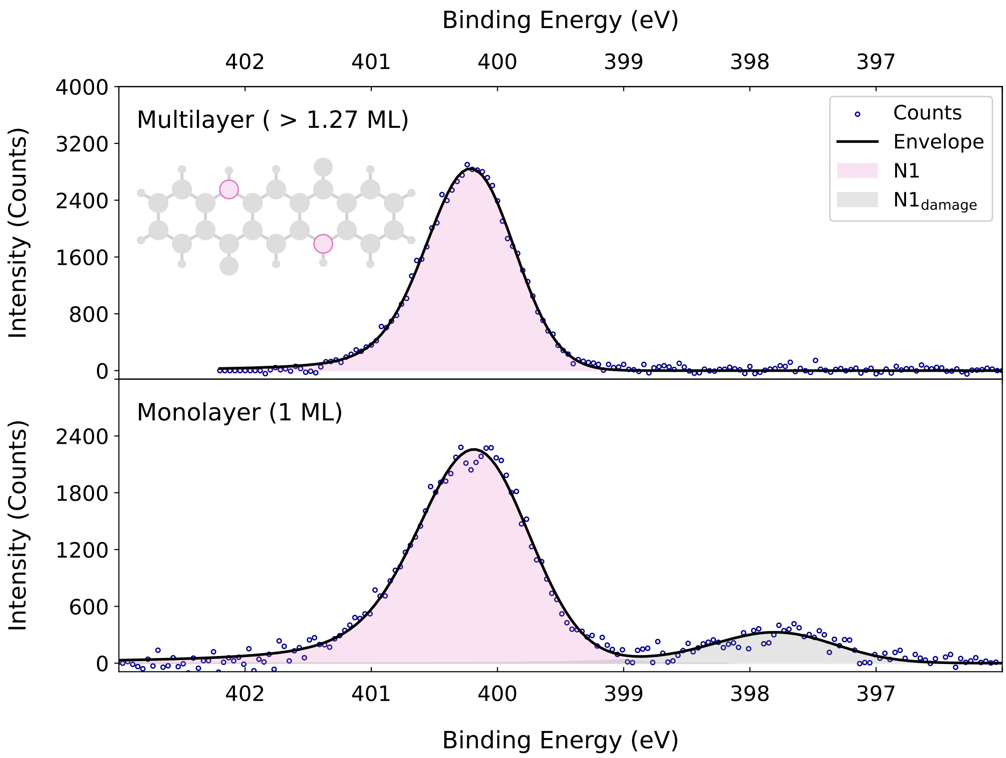


Figure 5.9: N1s **XPS!** spectrum for the α-phase of **QA!** on Ag(100) for a mutlilayer coverage compared to the **XPS!** spectrum for a coverage of 1 **ML!** with the experimental data points, the fitted components and the resulting envelope.

The **XPS!** spectrum with a multilayer coverage reveals a single component (N1 centered at a **BE!** of 400.15 eV, accompanied by a **FWHM!** of 0.82 eV. This observation signifies a subtle modification in the chemical environment surrounding the N atom. The peak associated with beam damage ($N1_{damage}$) remains undetected. This absence indicated that the spectrum of the multilayer originated predominantly from chemically intact nitrogen species. The **BE!** exhibits a slight increase (0.09 eV) in the multilayer, accompanied by a narrower **FWHM!**.

The parameters for the N1s **XPS!** spectrum with a multilayer coverage in **??** are listed in **??**.

Table 5.6: Fit parameters used in CasaXPS**CasaSoftwareLtd2022** for the multilayer α-phase of **QA!** on Ag(100) for the N1s photoemission lines.

peak	**BE!** / eV	area ratio	FWHM / eV
N1	400.15	2.00	0.82

5.2 C1s XPS spectra for the phase transition

The **??** illustrates C1s photoemission lines for **QA!** molecules absorbed on the Ag(100) surface, showing the transition from the α- to the β-phase with a coverage of 0.58 **ML!**. The **XPS!** spectra are arranged vertically, with the α-phase shown at the top, the β-phase at the bottom and a series of intermediate **XPS!** spectra representing the thermally induced phase transition. Maximum values for each individual peak are denoted by a colored vertical line. The phase transition occurs at a temperature of 450 K, which was the target temperature during the recording of the **XPS!** spectra. This temperature was chosen based on the previous results of the **LEED!** measurements, which indicated that the phase transition from the α- to the β-phase occurs at approximately 450 K.

In the α-phase, the C1s **XPS!** spectrum is characterized by distinct peaks corresponding to various chemically different C atoms, including the C_{arom}, C_{NH}, C_{CO} and C_{O} peaks, as discussed in **??**. In the final β-phase, the C1s **XPS!** spectrum demonstrates altered peak intensitites and **BE!** in comparison to the initial α-phase. The same fit model was used for both phases, but the **BE!** and **FWHM!** of the peaks were allowed to change during the fitting procedure.

It is noteworthy that all peaks undergo modification, which is indicative of a thermally induced structural rearrangement or reorientation of the **QA!** molecule. The sequence demonstrates a substantial trend for all peaks. A systematic, stepwise decrease in the **BE!** is exhibited by all four peaks. A comparative analysis of the C_{arom}, C_{CO} and C_{O} peaks reveals a consistent pattern of change.

A further difference in the change is observed in the C_{NH} peak. The shifts towards lower **BE!**s is less pronounced in comparison to the other peaks. This indicates that the environments of the other C1s photoemission lines, namely C_{arom}, C_{CO} and C_{O}, are changed in a similar way, while the C_{NH} C atoms experiences a different change in its chemical surrounding.

The analysis of the peak areas revealed that the integrated area of the peaks decreases by 13 % from the α- to the β-phase. It could be possible that this change is due to the different orientation of the **QA!** molecules in the β-phase, but this is not certain. Additionally, the number of **QA!** molecules in the β-phase may be different from the number of **QA!** molecules in the α-phase due to desorption of some **QA!** molecules during the thermal induced phase transition.

The time between the α-phase **XPS!** spectrum and the β-phase **XPS!** spectrum was 49 minutes. As the first **XPS!** spectrum of the α-phase was recorded, the temperature was set to 450 K, initiating the heating of the sample. The sample temperature was 450 K for the β-phase **XPS!** spectrum. The exact temperature for the intermediate **XPS!** spectra are not known.

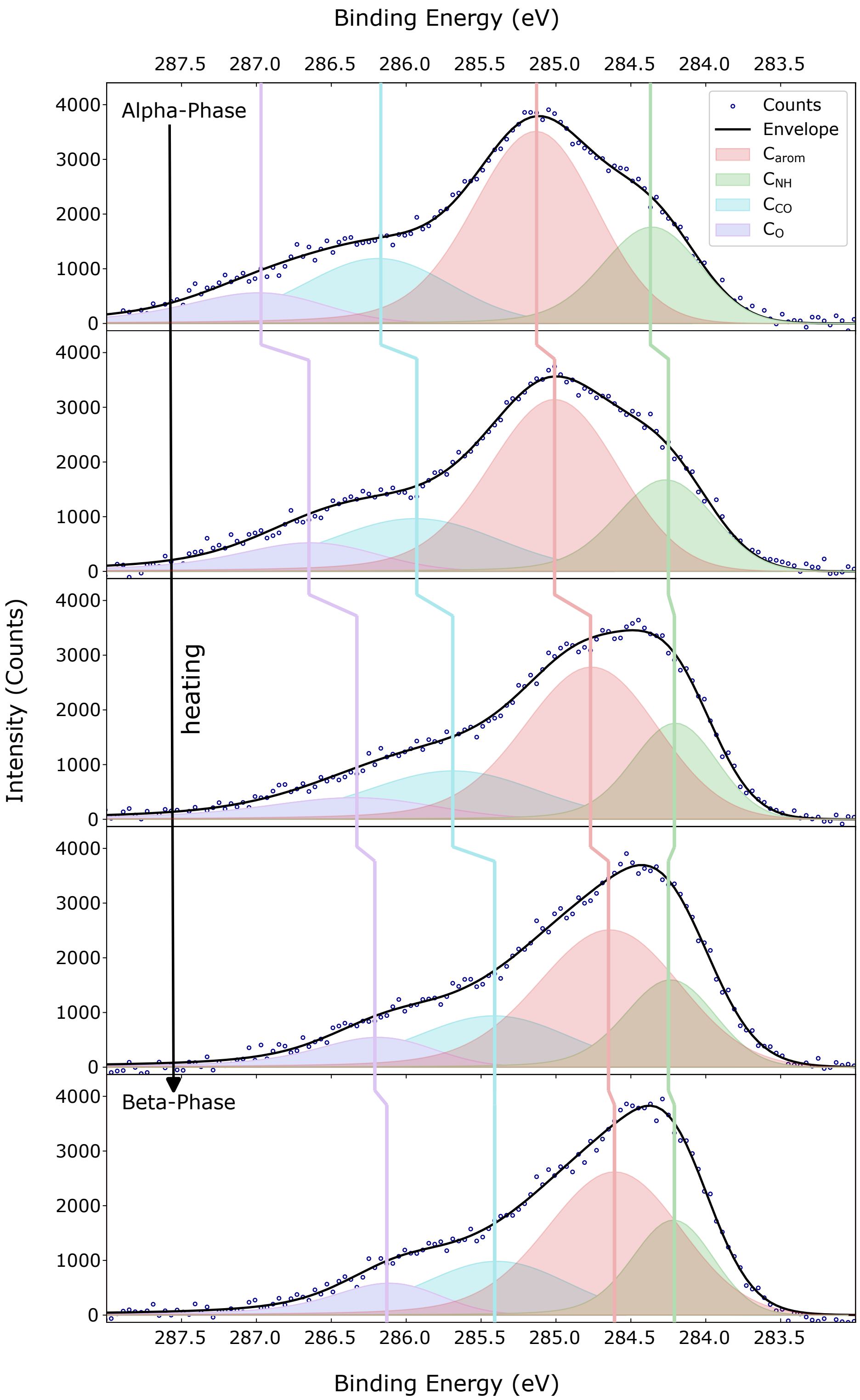


Figure 5.10: C1s **XPS!** spectrum for the phase transition from the α- to the β-phase of **QA!** on Ag(100) with a coverage of 0.58 **ML!**. Integrated area of the peaks decreases by 13 % from the α- to the β-phase. The vertical lines indicate the maximum of each peak.

5.3 The commensurate β-Phase

The ensuing discourse will examine the phase designated as the β-phase. The investigation of the β-phase was conducted once more on all three available photoemission lines: C1s, O1s, and N1s. To address this need, the present chapter has been meticulously subdivided into discrete sections, each dedicated to specific photoemission lines. A comprehensive analysis of a particular **XPS!** spectrum of the β-phase is presented in each section, along with a comparative analysis of the α- and β-phases.

5.3.1 C1s spectra

The C1s **XPS!** spectrum for the β-phase of **QA!** on Ag(100) is presented in **??**.

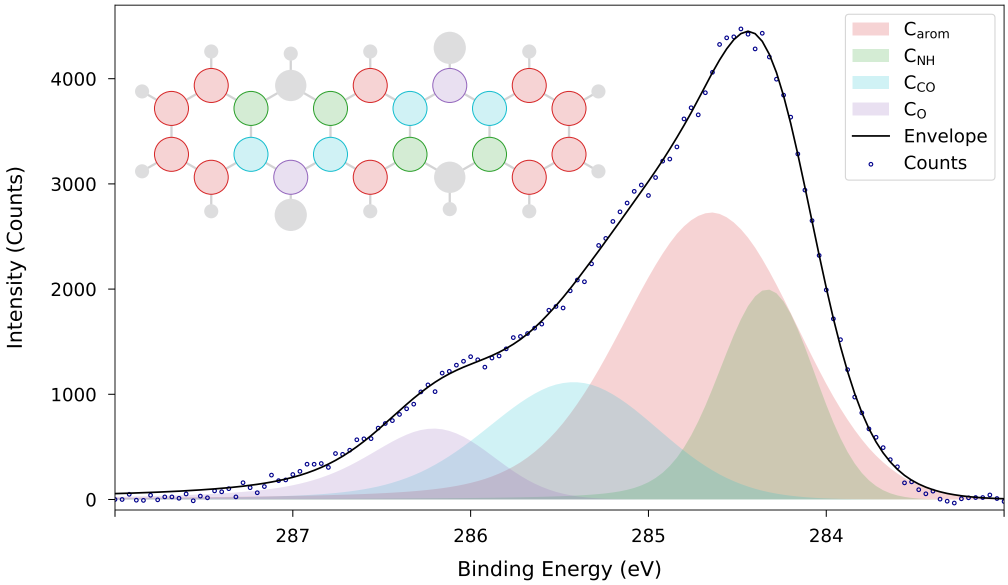


Figure 5.11: C1s **XPS!** spectrum for the β-phase of **QA!** on Ag(100) for a coverage of 0.54 **ML!** with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line).

The **XPS!** spectrum of **QA!** in the β-phase displays four distinct components, designated as C_{arom}, C_{NH}, C_{CO} and C_O, which bear the same nomenclature as in the α-phase. This finding suggests that the various C atoms present within the **QA!** molecule exhibit a persistent correlation with distinct peaks in the **XPS!** spectrum.

The primary component, C_{arom}, is located at a **BE!** of 284.55 eV with a **FWHM!** of 1.13 eV. The second peak, characterized by a **BE!** of 284.28 eV and a **FWHM!**

of 0.62 eV, is identified as the C_{NH} peak. The C_{CO} peak is located at a **BE!** of 285.32 eV, experiencing a **FWHM!** of 1.11 eV. In the vicinity of this peak, at a higher **BE!**, is the C_{O} peak at 286.07 eV with a **FWHM!** of 0.79 eV.

A qualitative investigation of the peak areas yielded analogous results to those of the α-phase, wherein the area ratios were set to the number of C atoms in the **QA!** molecule.

In summary, the **XPS!** spectrum for the C1s components of the β-phase offers significant insights into the chemical changes that occur between the α- and β-phase. The single peaks are compared to each other, thereby providing information about the local environment around the C atoms.

Table 5.7: Fit model of the β-phase for C1s.

peak	**BE!** / eV	area ratio	FWHM / eV
C_{arom}	284.55	10	1.13
C_{NH}	284.28	4	0.62
C_{CO}	285.32	4	1.11
C_{O}	286.07	2	0.79

In order to facilitate a more direct comparison of the β-phase to the α-phase, the C1s **XPS!** spectra of these two phases are presented in **??**.

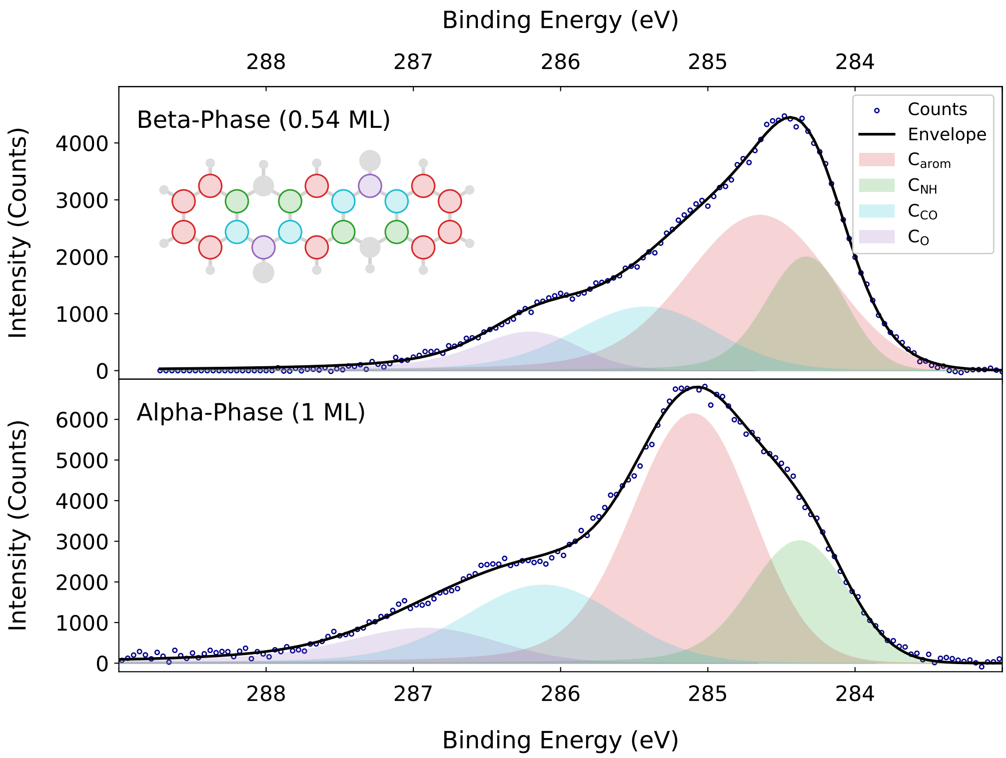


Figure 5.12: C1s **XPS!** spectrum for the α- and β-phase of **QA!** on Ag(100).

A close examination reveals that all four peaks shift to a lower **BE!**. The shift in the **BE!** of the C_{NH} peak is, by far, the least significant. The three other peaks, C_{arom}, C_{CO} and C_{O}, demonstrate a comparable shift in the **BE!**.

A further noteworthy discrepancy is the increase of the **FWHM!** for the C_{arom} peak, which is approximately 0.19 eV. The C_{NH} and C_{CO} peaks demonstrate a decrease in their **FWHM!** by approximately 0.1 eV. It is remarkable that the **FWHM!** of the C_{O} peak decreases by 0.37 eV. The observed variations of the **FWHM!** could be attributed to the interdependence of all four peaks.

In summary, the alterations in the **BE!** and the **FWHM!** values of the four peaks result in a significantly different **XPS!** spectrum. Consequently, **QA!** on Ag(100) in the β-phase is expected to demonstrate a significant change in its structure compared to the α-phase structure.

5.3.2 O1s spectra

In **??**, the **XPS!** spectrum for the O1s components for the β-phase of **QA!** on Ag(100) is presented.

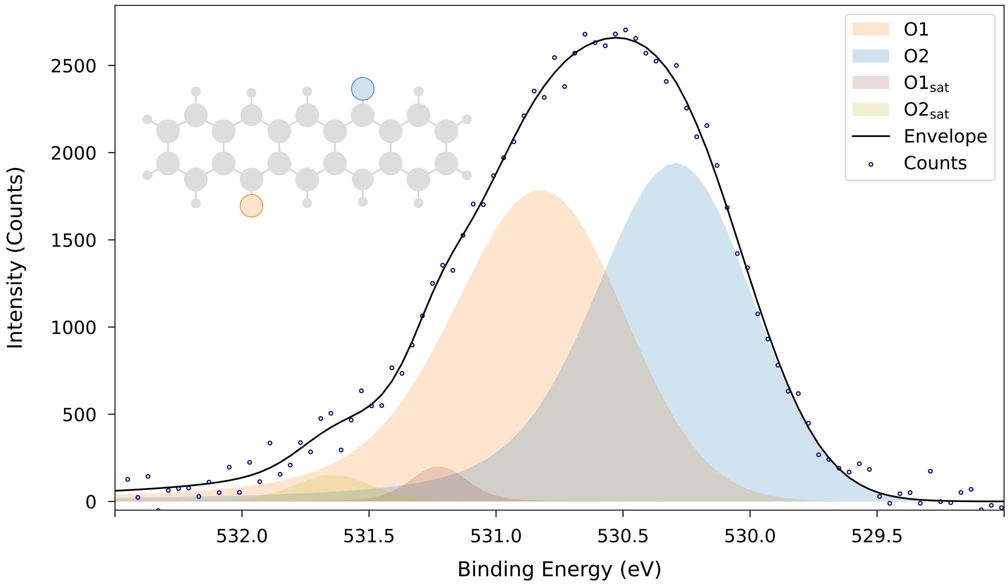


Figure 5.13: O1s **XPS!** spectrum for the β-phase of **QA!** on Ag(100) for a coverage of 0.54 **ML!** with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line).

The **XPS!** spectrum of the β-phase in **??** consists of two peaks, each with its own satellite. The detailed data of the overall four peaks are shown in **??**. The O1 peak is located at a **BE!** of 530.68 eV with a **FWHM!** of 0.71 eV. The respective satellite is at 531.66 eV with a **FWHM!** of 0.36 eV. This corrresponds to a shift in **BE!** of 0.98 eV.

The other peak, O2, has a slightly lower **BE!** of 530.19 eV but a similar **FWHM!** of 0.68 eV. The satellite $O2_{sat}$ also has a similar shift of 1.01 eV and is located at a **BE!** of 531.20 eV with a **FWHM!** of 0.45 eV. The O1 and O2 have an area ratio of 1:1, which corresponds to one O atom for each peak.

In summary, the **FWHM!** of two peaks are around 0.7 eV and both have a shake-up satellite with a higher **BE!** by approximately 1.0 eV. The two peaks are 0.5 eV apart from each other.

Table 5.8: Fit model of the β-phase for O1s.

peak	**BE!** / eV	area ratio	FWHM / eV
O1	530.68	1	0.71
$O1_{sat}$	531.20	0.14	0.45
O2	530.19	1	0.68
$O2_{sat}$	531.66	0.05	0.36

To enable a better comparison between α- and β-phase, both spectra are depicted in **??**.

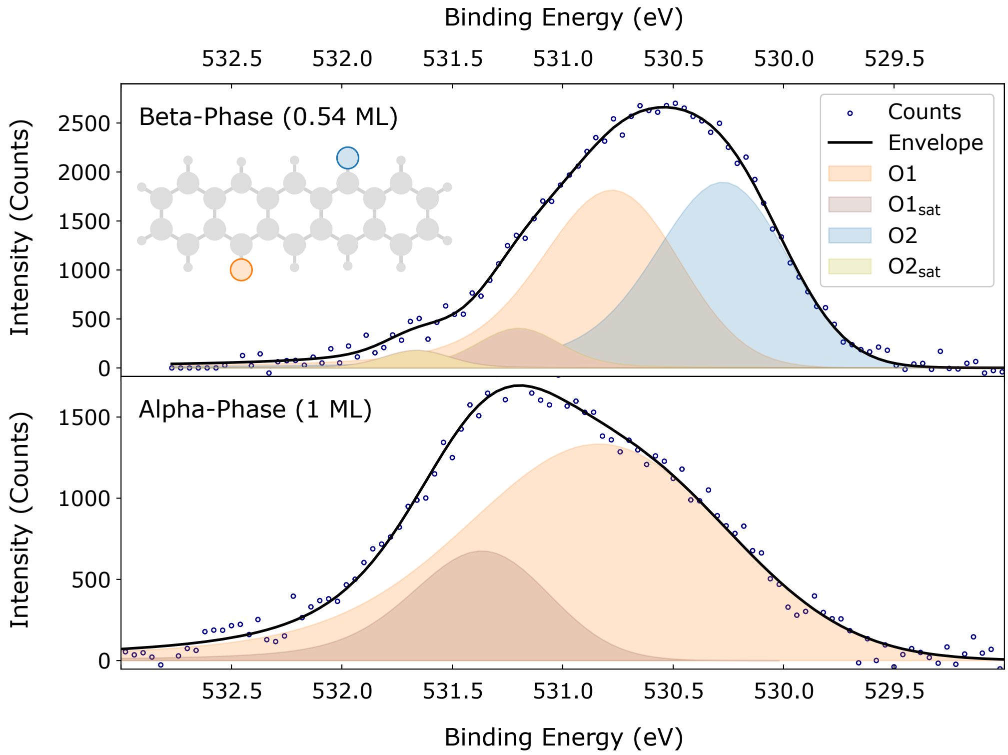


Figure 5.14: O1s **XPS!** spectrum for the α- and β-phase of **QA!** on Ag(100).

The investigation of the differences between the two phases yields that the O1 peak does not change its position but exhibits a change in its **FWHM!** towards a smaller value. Same holds true for the $O1_{sat}$ peak, which also changes its **BE!** by approximately 0.4 eV.

The most significant change is the appearance of a second peak for the β-phase. This means that the two chemically equivalent peaks from the α-phase have changed their environment, leading to two chemically different O atoms in the β-phase.

5.3.3 N1s spectra

In **??**, the N1s **XPS!** spectrum for the β-phase of **QA!** on Ag(100) is presented.

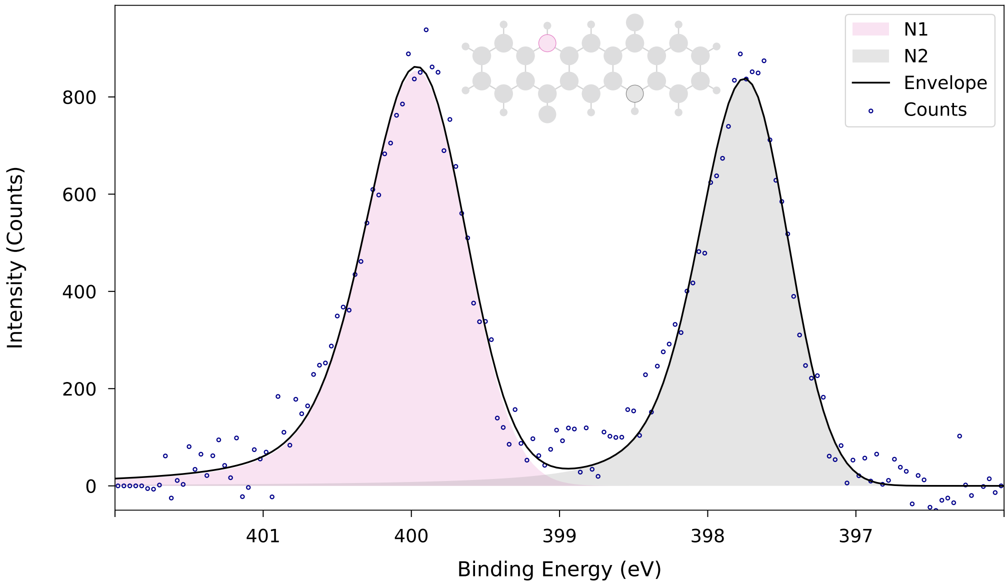


Figure 5.15: N1s **XPS!** spectrum for the β-phase of **QA!** on Ag(100) for a coverage of 0.54 **ML!** with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line).

?? and **??** enable a quantitative description of the N1s **XPS!** spectrum for the β-phase. As seen, the β-phase has two different peaks for N1s, namely N1 and N2. Both peaks have a similar **FWHM!** of 0.77 eV and 0.69 eV. Furthermore, they have the same intensitiies and an area-ratio of 1:1. This indicates that each peak corresponds to one of the N atoms of **QA!**.

The main difference of the two peaks is their **BE!**. The N1 peak is located at a **BE!** of 399.87 eV while the N2 peak has a **BE!** of 397.66 eV. This means that the peaks are separated by 2.20 eV. In comparison to the shifts in **BE!**s for the other photoemission lines, this value seems to indicate a quite drastic change in the chemical environment of one of the N atoms.

Table 5.9: Fit model of the β-phase for N1s.

peak	**BE!** / eV	area ratio	FWHM / eV
N1	399.87	1	0.77
N2	397.66	1	0.69

To enable a comparative analysis between the α- and the β-phase, the two phases are depicted in a juxtaposed arrangement in **??**.

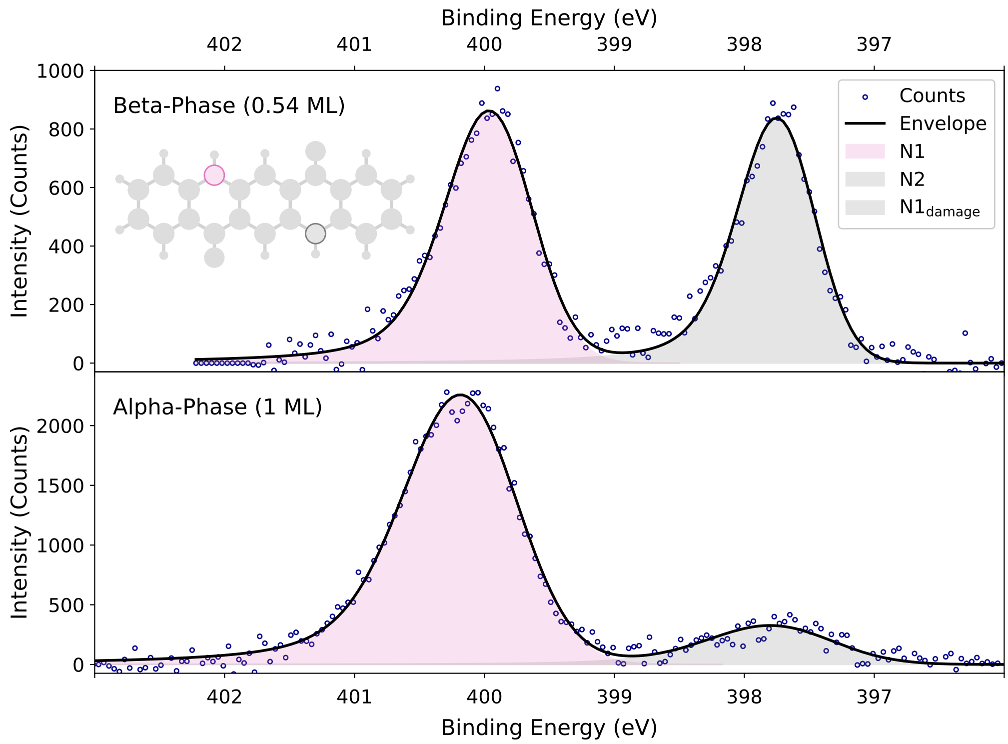


Figure 5.16: N1s **XPS!** spectrum for the α- and β-phase of **QA!** on Ag(100).

The comparison of the α- and the β-phase yields interesting results about the N atoms. First, the N1 peak keeps almost the same **BE!** in both phases. There is only a change in **BE!** of about 0.188 eV towards a lower value. The more significant change can be observed in the change of the **FWHM!**, decreasing from 1.017 eV to 0.774 eV. This corresponds to a sharper peak in the **XPS!** spectrum, indicating a more homogenous environment for the N1 peak.

Nevertheless, the other peak, namely N2, reveals a more intriguing finding, prompting further examination. The N2 peak has exactly the same **BE!** as the $N1_{damage}$ peak of the α-phase. As seen in **??**, the areas of the second peak in the different phases can not be compared anyhow, which makes an comparison of the **FWHM!** unsuitable, because the $N1_{damage}$ peak is due to beam damage and the N2 peak are a chemically different N atoms in the β-phase.

5.4 Binding energy changes upon the phase transition

For a further comparison of the differences in the **BE!**s between the α- and the β-phase, an overview of these are illustrated in **??**.

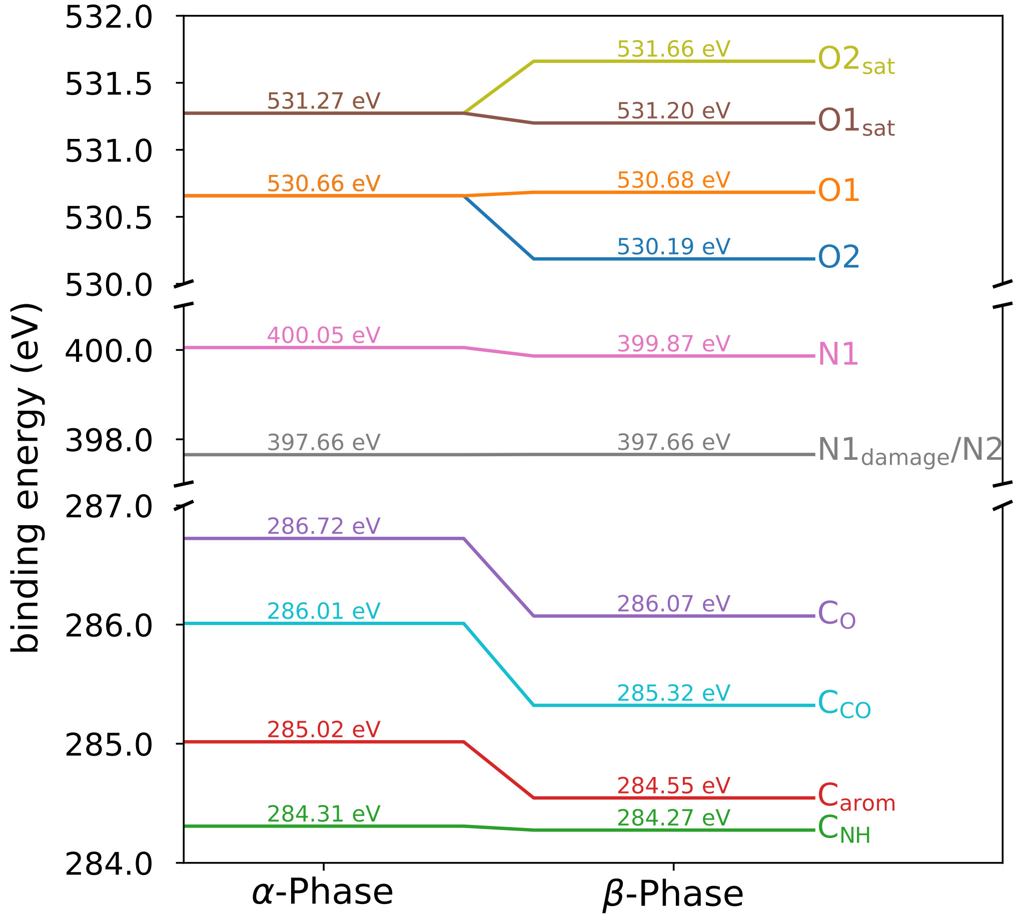


Figure 5.17: Illustration of the shift in the **BE!** from the α- to the β-phase of **QA!** on Ag(100) for the different peaks in the **XPS!** spectra for all core-levels.

Initially, a significant trend is observed in all peaks across all **XPS!** spectra. The **BE!**s of all atoms in the β-phase are lower than or similar to those of the atoms in the α-phase.

Starting with the **XPS!** spectra for C1s, the differences in the **BE!**s are subsequently examined in detail. Shifts in the **BE!** for three of the four peaks, namely C_{arom}, C_{CO} and C_O are similar. They all decrease by 0.5-0.7 eV. Conversely, the **BE!** of the C_{NH}

peak does not significantly decrease. This indicates that the C atoms next to the amine undergo a different change during phase transition than the other C atoms.

Consequently, the **XPS!** spectra for N1s are of great interest. As previously mentioned, the β-phase exhibits two peaks instead of one for N1s. The second peak in the β-phase is more than 2 eV lower in **BE!** than the first peak. This represents a significant difference in the **BE!** that is not observed for any other peak. To provide an example, the **BE!** difference between the C_{NH} and the C_O peak, despite their very different chemical environments, is only 1.8 eV.

Lastly, the **BE!**s of the two phases for the O1s **XPS!** spectra are compared. Again, there are two distinct peaks in the β-phase instead of one in the α-phase. The **BE!** of one peak remains at roughly the same value, while the other decreases by about 0.4 eV. This result is similar to the change in the **BE!** of individual components observed in the C1s **XPS!** spectra.

6 Discussion

The α- and β-phase of **QA!** on the Ag(100) surface were investigated using **XPS!** for three different components: C1s, O1s and N1s. The following discourse will examine the results obtained from **??**, which will be divided in the same manner as the results section.

6.1 The α-Phase

The C1s **XPS!** spectra consist of four distinct components, corresponding to the distinct types of C atoms in the **QA!** molecule, namely C_{arom}, C_{NH}, C_{CO} and C_{O}.

The C atoms C_{arom} have the least influence from the functional groups. This leads to a **BE!** in the center of the spectrum. Due to the number of C atoms associated with this peak, the peak area is the most substantial, and the peak provides the primary contribution to the overall shape of the spectrum. The **BE!** of the peak appears to be realistic, as evidenced by the agreement of the **BE!** with that of other molecular adsorbates on the Ag(100) surface, such as **PTCDA!** (**PTCDA!**) or indigo. For **PTCDA!**, the **BE!** of the C atoms inside the ring is 284.91 eV,[Bauer2014] which is very similar to the **BE!** of the C_{arom} peak (285.02 eV) for **QA!**. In addition, the **BE!** of the corresponding peak in indigo (285.5 eV)[Brunet2011, Michels2021] is also very similar to the **BE!** of the C_{arom} peak in **QA!**.

The C_{CO} peak is located at a higher **BE!** than the C_{arom} peak, which is expected due to the electronegative O atom. The **BE!** of the C_{CO} peak (286.01 eV) is also in line with the literature, as it is similar to the **BE!** of C atoms next to a carbonyl groups in other molecular adsorbates, such as **PTCDA!** (285.47 eV).[Bauer2014]

The C_{O} peak is located at an even higher **BE!** than the C_{CO} peak, which is also expected due to the electronegative O atom. The **BE!** of the C_{O} peak (286.72 eV) also matches to the literature, as it is similar but not as high as the **BE!** of carbonyl C atoms in **PTCDA!** (288.92 eV)[Bauer2014] or in indigo (287.5 eV).[Brunet2011, Michels2021]

The peak corresponding to the C atoms adjacent to the amine, designated as C_{NH}, is located at a slightly decreased **BE!** in comparison to the C_{arom} peak. This behavior of aromatic systems with amine functional groups inside, such as those found in pyridine, is typically observed.[Mendolicchio2019, Bagus2025] In contrast to this, the **BE!** of the C_{NH} peak in indigo (285.9 eV)[Brunet2011] is higher than the **BE!** of the C_{arom} peak, which is not the case for **QA!**. Nevertheless, the structures and the bonding situation of the two molecules are different, which could explain the different **BE!**s.

The **XPS!** spectra for the N1s component exhibit a single peak in the α-phase. The **BE!** of the N1s peak (400.0 eV) is in line with the literature, as it is similar to the **BE!** of N atoms in other molecular adsorbates, such as indigo (399.9 eV).[**Xu2024**]

The peak that arises due to beam damage, $N1_{damage}$, has a shift towards smaller **BE!**s of around 2.20 eV. This large difference indicates a drastic change in the chemical environment of the N atoms. Compared to other shifts in the **BE!**, e.g. for the C atoms, the influence of other neighboring atoms are not enough to explain this change. For this reason, something different has to occur at the N atom due to beam damage.

A typical observed phenomena for the amine group is the deprotonation of the N atom. The deprotonation of amine groups in **XPS!** spectra was examined with a shift in the **BE!** of about 2-3 eV for glyphosate, molecules similar to pyridine and polyaniline films.[**Ruano2021**, **Kang1990**, **M.Mahat2015**] This matches to the results found for the $N1_{damage}$ peak, leading to the assumption that the x-ray beam deprotonates some of the N atoms.

Last, the **XPS!** spectra for the O1s components are discussed. The **BE!** of the O1 peak (530.66 eV) matches similar systems. e.g. **PTCDA!** on the Ag(100) surface (531.93 eV)[**Bauer2014**] or indigo (530.2 eV).[**Xu2024**]

A comparable satellite peak was also measured for **PTCDA!** with a difference to the main component of 2.28 eV.[**Bauer2014**] The satellite peaks for the two adsorbates have similar shifts in their **BE!**s and similar area ratios with respect to the main peak. This indicates that the results of the O1s **XPS!** spectra and the fit model is realistic. Indigo exhibits a similar satellite peak, but with a shift of 1.0 eV.[**Xu2024**]

Nevertheless, the photoemission lines for the O1s components should also exhibit a second peak, which corresponds to the O atom that cannot form a hydrogen bond with the amine group of another molecule because some of the N atoms are deprotonated due to beam damage. This peak is not observed in the **XPS!** spectra, which confirms that the number of deprotonated N atoms is low and does not have a significant impact on the overall spectra for the α-phase. Nevertheless, a second peak for the O1s component could be hidden in the **XPS!** spectra, increasing the **FWHM!** of the O1 peak.

In summary, the entirety of the peaks in the **XPS!** spectra adequately describe the molecular structure of **QA!**. The **BE!** of the different photoemission lines is consistent with the given values from the literature. In addition, the observed peaks in the **XPS!** spectra corroborate the existing structure model proposed by N. Humberg.[**Humberg2024**] The chemical environments of all N and O atoms are similar within the structure model, because all O atoms of one molecule form a hydrogen bond with the amine group of another molecule. This is evident from the **XPS!** spectra, which exhibit only a single peak for the corresponding atoms. Furthermore, the results indicate that all molecules in the structure have a similar chemical environment, which also matches to the **POL!** type structure model.

6.2 Phase transition

The examination of the phase transition was done by measuring a sequence of C1s **XPS!** spectra while increasing the sample temperature. This sequence shows in principle that the phase transition from the α- to the β-phase occurs in a stepwise manner while heating the sample to 450 K. A close examination of the data reveals a transition in each peak from one spectrum to the next **XPS!** spectrum. This indicates that all molecules would undergo simultaneous and analogous changes.

While this assumption may be theoretically valid, it appears to be improbable that a stepwise change occurs. A more probable theory posits that the phase occurs for some molecules prior to others, resulting in a decreases of the α-phase and an increase the β-phase. This observation aligns with the findings reported by N. Humberg.[Humberg2024] Nevertheless, the exact process of the phase transition can not be verified with the given data, as a superposition of the two phases could not be tested.

6.3 The commensurate β-Phase

For the β-phase of **QA!** on the Ag(100) surface one set of **XPS!** spectra for all photoemission lines were recorded. First, the C1s **XPS!** spectra are discussed. The β-phase exhibits four distinct peaks, that correspond to the same C atoms as in the α-phase, having the same relative positions, with the C_{arom} in the center, the C_{NH} peak towards lower, the C_{CO}- and the C_O peak towards higher **BE!**s. The three peaks, C_{arom}, C_{CO} and C_O, have a similar shift towards lower **BE!** upon phase transition in their **BE!**s. This corresponds to similar changes in their chemical environment.

The lower **BE!** could be explained with stronger adsorbate-substrate interactions. Thereby, the metal surface donates electron density to the molecule, resulting in lower **BE!**s. This process is also observed in literature and thus a valid explaination for the observed change in the **BE!**s.[Moulder1993]

However, the change exhibited in the C_{NH} peak is distinct. The **BE!** remains constant during the phase transition; nevertheless, this phenomenon should not be interpreted in such a simplistic manner. Instead, it should be interpreted as a similar change in the **BE!** towards lower values due to a stronger interaction with the surface upon phase transition and then another effect that increases the **BE!** again. The phase transition involves a change in the N atom bound to the C_{NH} atom, as shown in the N1s **XPS!** spectra. The bond between the C and N atoms has a significant impact on the electron density at the C_{NH} atom and could decrease the electron density. This decreased electron density results in an increase in the **BE!** of the C atom.

It is noteworthy, that the C atoms in the C_{NH} peak are directly next to the N atoms. For the β-phase, two different nitrogen peaks are present, which should also lead to two different C_{NH} peaks. However, the C_{NH} peak could not be resolved into two different peaks. The same applies to the C atoms next to the O atoms, as the O1s **XPS!** spectra also exhibit two distinct peaks.

This leads to the subsequent discussion of the N1s **XPS!** spectra for the β-phase, where the one peak for both N atoms from the α-phase divides into two different peaks. At the outset, the **BE!** of the N1 peak undergoes a slight decrease upon the phase transition. This decrease is less significant than that observed for the other atoms. This implies that the N atom, corresponding to the N1 peak, exhibits a marginal increase in electron density, while the overall changes are analogous to those observed in the entire molecule. The observed increase in electron density may be attributed to enhanced adsorbate-substrate interactions.

The second peak for the β-phase is precisely coincident with the position of the $N1_{damage}$ peak from the α-phase, which corresponds to beam damage. It can be supposed that the N atom corresponding to the N2 peak has the equivalent chemical environment as the $N1_{damage}$ peak. As previously stated, the $N1_{damage}$ peak corresponds to a deprotonated N atom, a finding that aligns with the literature.[**Kang1990**, **Ruano2021**, **M.Mahat2015**] This observation suggests the hypothesis that the observed N2 peak corresponds to a deprotonated N atom as well.

This means that in each **QA!** molecule exactly one of the two amine groups is deprotonated due to the phase transition. The energy, needed for the deprotonation, has to be overcompensated by the different orientation and stronger adsorbate-substrate interaction in the β-phase, because the β-phase is thermodynamically favored.[**Humberg2024**] It is conceivable that the deprotonated N atoms bond to the Ag(100) surface and donate electron density to the metal, which would lead to a stabilization of the β-phase. This process could then overcompensate the energy loss due to the deprotonation and the loss of one hydrogen bond. Furthermore, the protons could interact with the Ag(100) surface and catalytically form hydrogen molecules (H_2). This process could be observable by measuring the mass of H_2 with a mass spectrometer upon the phase transition.

The transition observed from a single peak in the α-phase to two peaks in the β-phase was likewise detected in the O1s **XPS!** spectra. The **BE!** of the O1 peak remains nearly constant. This phenomena is analogous to that observed for the N1 peak.

Consequently, it can be deduced that the N atom of the N1 peak and the O atom of the O1 peak are the two atoms that interact with each other in the α- and β-phase. These two atoms still have a hydrogen bond in the β-phase between the keto and the amine group, which leads to similar **BE!**s for both phases.

The second peak in the O1s **XPS!** spectrum for the β-phase then corresponds to the other O atom that interacts with the N2 peak. This O2 peak exhibits a lower **BE!** in comparison to the O1 peak by 0.5 eV. This corresponds to an increased electron density at the O atom, which could be result of a missing hydrogen bond between the O atom from one molecule and the hydrogen atom of an amine group from another molecule. The O atom then has a higher electron density, which leads to a lower **BE!**.

For the β-phase, both O1s peaks in the **XPS!** spectra have satellites. These are at 0.6 eV and 1.4 eV higher **BE!**s than the main peaks, which corresponds to a similar **BE!** for the $O1_{sat}$ peak and an increase in the **BE!** of the $O2_{sat}$ peak compared to the α-phase. This indicates that the shake-up process also changes between the two phases.

The experimental results from the **XPS!** spectra have to be compared to the structure model from N. Humberg.[Humberg2024] First, the structure of the β-phase is commensurate. This is also seen in the C1s **XPS!** spectra, because it shows precise positions for each of the C atoms. Altogether, the C1s **XPS!** spectra match to the structure model of the β-phase.

However, the O1s and N1s **XPS!** spectra contradict the structure model. The **XPS!** spectra show for both atom types two distinct peak with a 1 : 1 area ratio. This observation does not align with the structure model. According to N. Humberg,[Humberg2024] the structure of the β-phase has 1.5 hydrogen bonds per **QA!** molecule, which would correspond to two distinct peaks with an area ratio of 1.5 : 0.5 (3 : 1).

Apart from this, the change in the **BE!** for the N1s **XPS!** spectra indicates that 1 of the two N atoms per molecule is deprotonated. This finding was not taken into account for the structure model. Furthermore, the deprotonation of one of the two N atoms would lead to the fact, that 1.5 hydrogen bonds per molecule are not possible anymore.

Consequently, the results of the experiments lead to a different structure model for the β-phase. Therefore, two theoretically possible new structure models are depicted in **??**, which match to the SPA-LEED image and STM results from N. Humberg.[Humberg2024] The structure models will be discussed in the following.

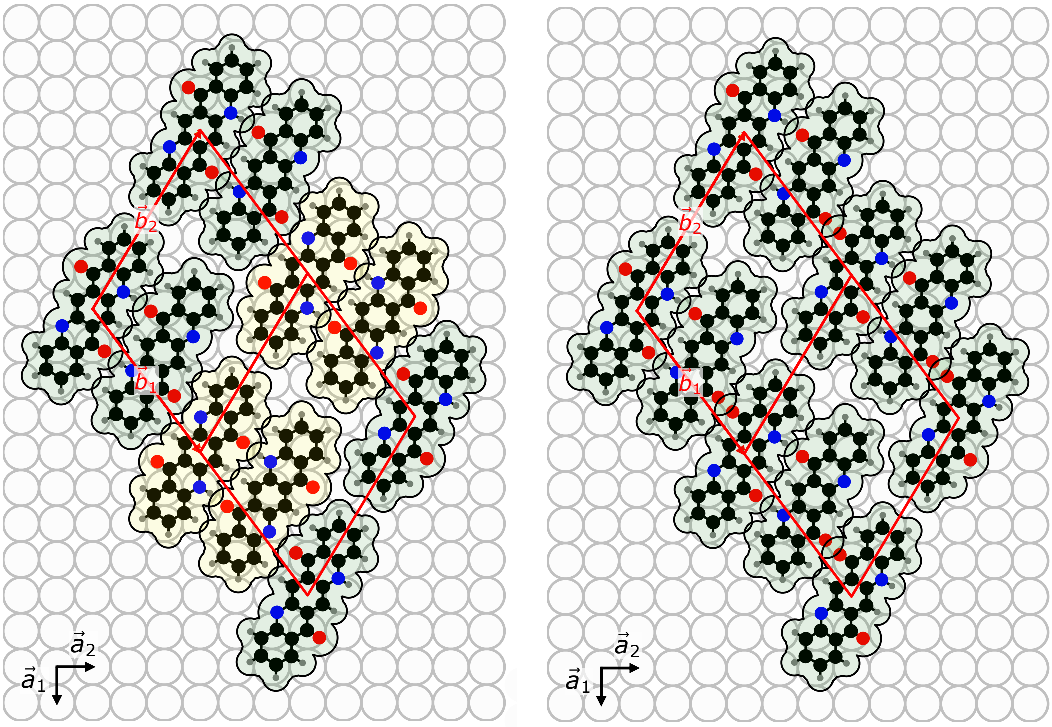
The first structure model is similar to the one of N. Humberg.[Humberg2024] The positions of the molecules are exactly the same and dimers are also formed. The most important change is the deprotonation of one of the N atoms per molecule. The most probable N atom for the deprotonation is the one that is not in between the molecules of the dimer, otherwise there would be no reason to form these dimers. In addition, the high electron density at the deprotonated N atom and at the O atom would repel each other. This results in two hydrogen bonds between the two molecules of the dimer, meaning one hydrogen bond per molecule. No further hydrogen bonds are formed in this structure.

The structure seems the preferred one but there are further aspects that need to be discussed. In model 1, the deprotonated N atom, which then has a high electron density, is opposite to the O atom, which also has a high electron density. Usually, the negative charges should repel each other. Nevertheless, it is possible that the N and the O atom change their height and approach the surface. This could increase the instance between the two atoms. Furthermore, both atoms could donate electron density to the Ag(100) surface, which should also decrease the repulsion.

Another aspect to consider are the chemical environments of the O atoms without hydrogen bonds. There are two different chemical environments: one environment is characterized by a deprotonated amine group opposite to the O atom in the next molecule, while the other environment is opposite to the most outer ring of the **QA!** molecule. This difference in chemical environments could lead to a changed electron density in the O atoms, which would then result in different **BE!**s in the **XPS!** spectra for these two environments. This was not observed in the **XPS!** spectra. A possible explanation for this could be that the chemical environments are not sufficiently different to result in a measurable change in the **BE!**, as the difference is only in the neighboring molecule and not in the molecule itself.

The second structure model is characterized by more changes with respect to the one of N. Humberg.[**Humberg2024**] Instead of the heterochiral dimers, the structure model 2 is homochiral. This means that the molecules do not have to rotate on the surface for the phase transition from the α- to the β-phase. The model also forms the same amount of hydrogen bonds as model 1.

However, one significant issue is identified with model 2. As depicted in the structure model, the O atom that is not involved in hydrogen bonding of one molecule is directly opposite to the O atom of another molecule. The plausibility of this occurrence appears to be minimal, because they should repel each other and the distance between them is so small that their van-der-Waals radii overlap. Nevertheless, it is possible that both O atoms approach the surface and donate their electron density to the surface. This could increase the distance between the O atoms and decrease the repulsion. Consequently, model 2 appears to be less probable than model 1 for the reasons previously enumerated.



(a) structure model 1

(b) structure model 2

Figure 6.1: Two different structure models for the β-phase of **QA!** on Ag(100).

7 Summary and Outlook

This study investigated the binding energies of the different atomic species and the consequential molecular orientation of **QA!** in different phases adsorbed on the Ag(100) surface using **XPS!**. The primary research question addressed how the **QA!** molecules interact with the substrate and what changes in its binding situation occur upon the phase transition. Specifically, the aim was to understand the intermolecular interactions via hydrogen bonds and the adsorbate-substrate.

The **XPS!** measurements revealed distinct peaks for the C, N and O atoms with different chemical environments within **QA!**. The photoemission lines for C1s show different peaks for aromatic C atoms (C_{arom}), C atoms bonded amine group (C_{NH}), C atoms in carbonyl groups (C_{CO}) and C atoms bonded to oxygen (C_{O}). All these peaks were observed in both the α- and the β-phase. Furthermore, the different peaks in the C1s **XPS!** spectra exhibit a shift towards lower **BE!**s between the two phases, indicating an increased interaction with the substrate in the β-phase.

The N1s and O1s **XPS!** spectra show one peak for the α-phase, whereas the β-phase exhibits two distinct peaks for the N and O atoms with an area ratio of 1 : 1. This indicates a change in the number of hydrogen bonds per molecule between the two phases, with the α-phase having two hydrogen bonds per molecule and the β-phase having one hydrogen bond per molecule. Additionally, the consequence of beam damage due to exposure of **QA!** to the x-ray was investigated, demonstrating the deprotonation of the amine group as a result of this exposure.

As already stated, the N1s **XPS!** spectra show a peak for the deprotonated N atoms in the β-phase, which is not present in the α-phase. This indicates that the deprotonation of one N atom per molecule occurs during the phase transition from the α- to the β-phase, explaining the change in the number of hydrogen bonds per molecule.

The results for the α-phase are consistent with the prior findings by N. Humberg.[**Humberg2024**, **Humberg2020**] The proposed structure model has been confirmed and the **XPS!** spectra for the C1s, N1s and O1s components can be subdivided into multiple peaks, which can be allocated to particular atoms of **QA!**.

The subsequent examination of the results for the β-phase reveals inconsistencies with the proposed structure model by N. Humberg.[**Humberg2024**, **Humberg2020**] It was ascertained that the β-phase is composed of two distinct O atoms, exhibiting a stoichiometric ratio of 1 : 1. A similar observation was made in the case of the N atoms. The observed shifts in the **BE!** are indicative of the deprotonation of one of the N atoms. This results in the formation of a single hydrogen bond per molecule.

The findings of this study have led to the proposition of two novel structure model in **??**.

Nevertheless, the extant structure models leave unresolved questions. To further refine our comprehension of **QA!** on the Ag(100) surface, it is imperative to undertake additional research to develop a more detailed structure model. **NIXSW!** (**NIXSW!**) measurements could be used to determine the adsorption height of the atoms, which allow a revision of the structure models. Beyond that, different **NIXSW!** measurements could be used for triangulation of the β-phase. This should give exact adsorption sites for the atoms and allow a detailed structure model.

8 Appendix

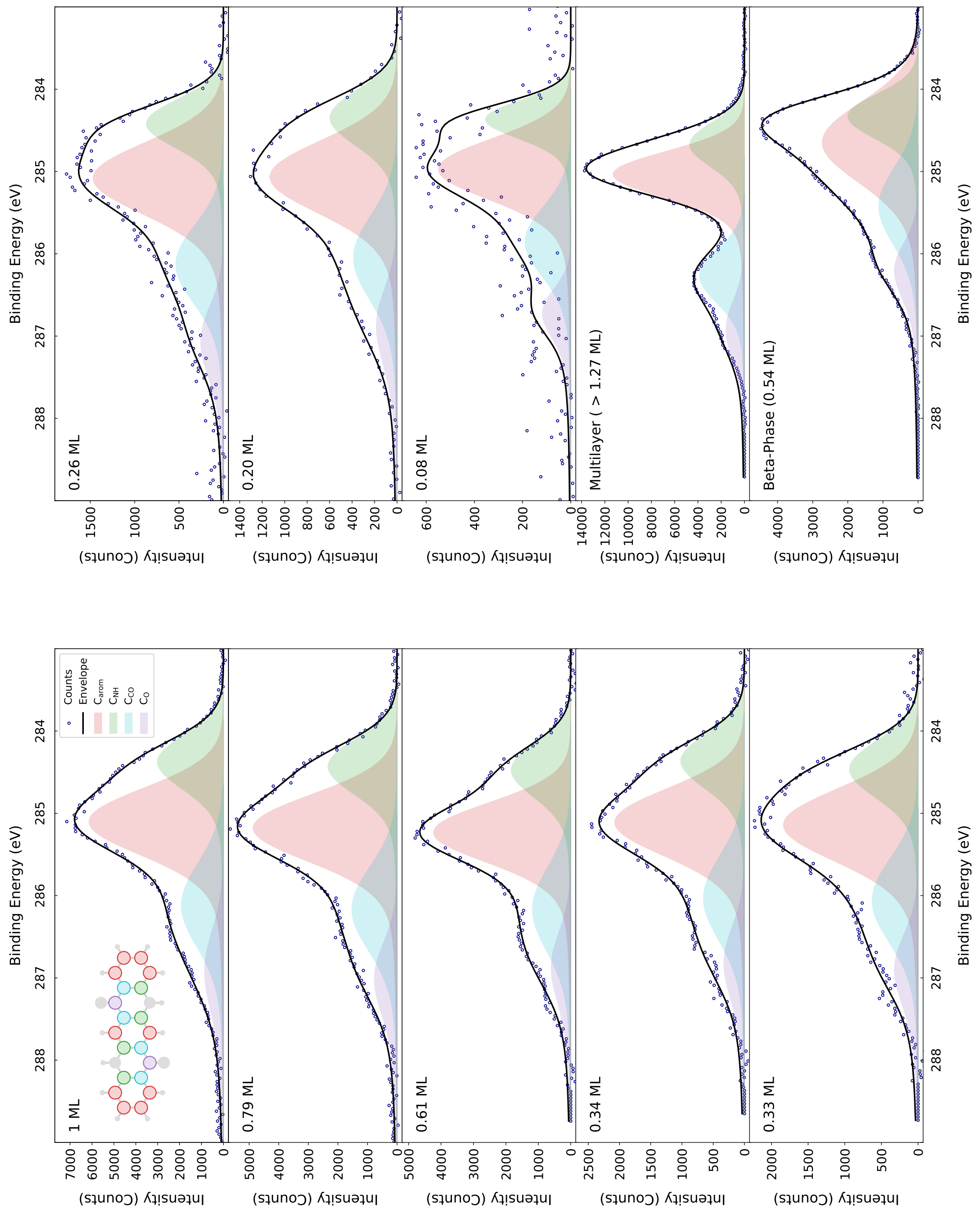


Figure 8.1: Collection of all C1s core-level **XPS!** spectra with the different fitted peaks of all performed experiments. Each spectrum corresponds to a different preperation. The coverage of the first spectrum is set to 1 **ML!**.

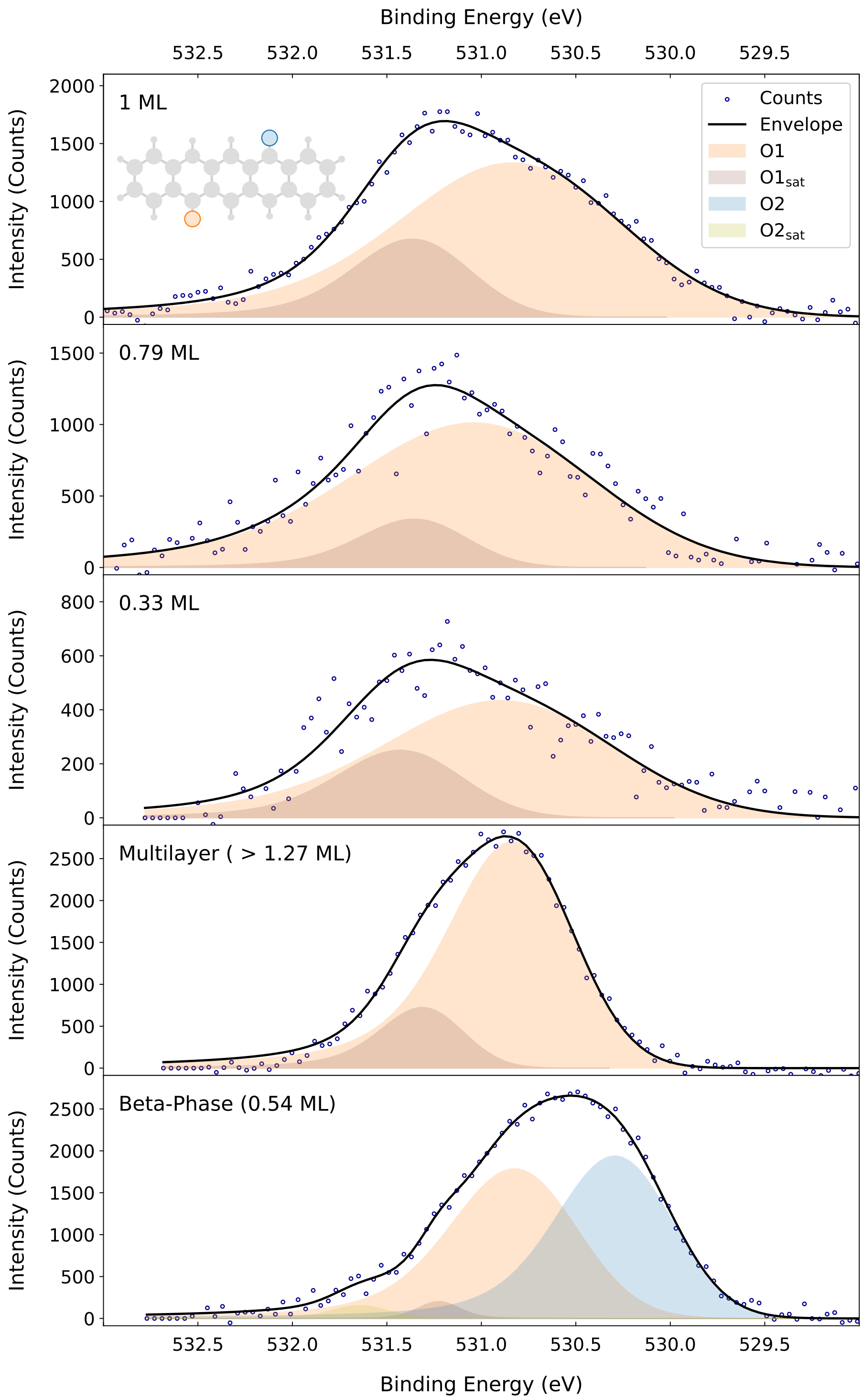


Figure 8.2: Collection of all O1s core-level **XPS!** spectra with the different fitted peaks of all performed experiments. Each spectrum corresponds to a different preperation. The coverage of the first spectrum is set to 1 **ML!**.

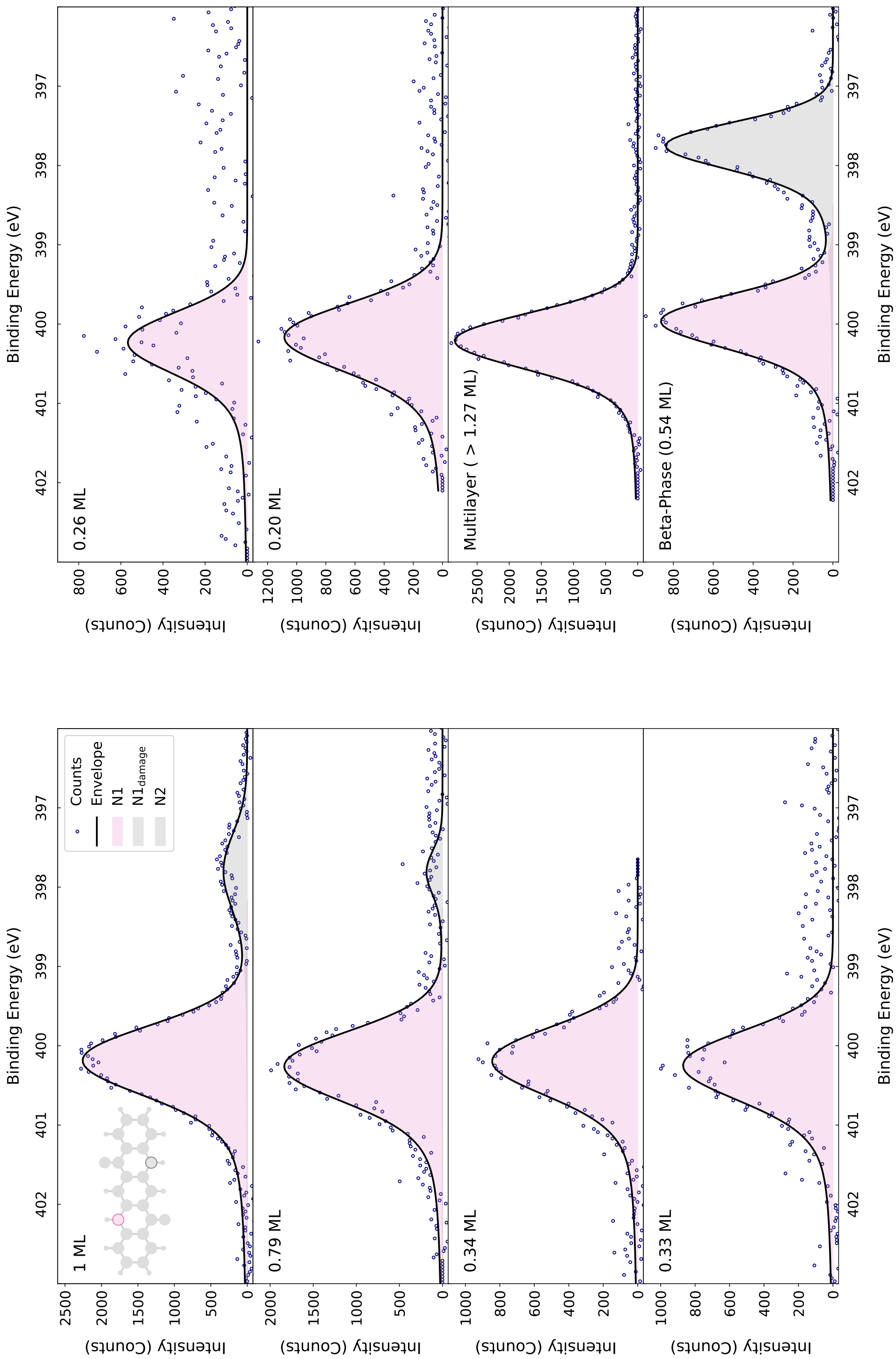
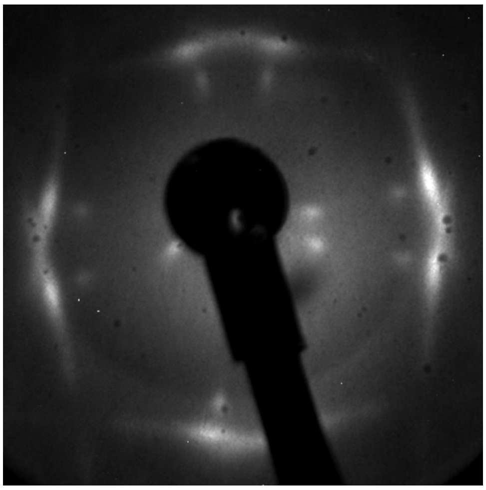
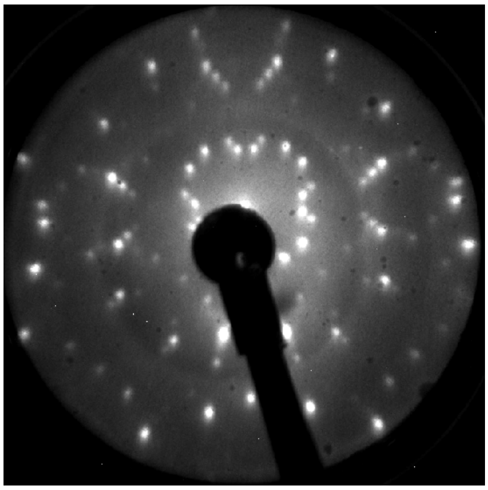


Figure 8.3: Collection of all N1s core-level **XPS!** spectra with the different fitted peaks of all performed experiments. Each spectrum corresponds to a different preperation. The coverage of the first spectrum is set to 1 **ML!**.



(a) **POL!** α-phase



(b) Commensurate β-phase

Figure 8.4: **LEED!** patterns for the α- and the β-phase at 25.5 eV taken at the I09 beamline of the Diamond Light Source.

List of abbreviations

BE binding energy

fcc face-centered cubic

FWHM full width at half maximum

LEED low energy electron diffraction

ML monolayer

NIXSW normal incidence x-ray standing wavefield absorption

OFET organic field-effect transistor

OLED organic light-emitting diode

OPV organic photovoltaic

POL point-on-line

PTCDA perylenetetracarboxylic dianhydride

QA quinacridone

SPA-LEED spot profile analysis low energy electron diffraction

STM scanning tunneling microscopy

UHV ultra-high vacuum

XPS x-ray photoelectron spectroscopy